

Equivalence of Scott’s Three Presentations of Domain Theory

Lars Warren Ericson¹

¹Catskills Research Company

¹https://github.com/catskillsresearch/scott_models

¹lars.ericson@catskillsresearch.com

August 25, 2026

ORCID: 0000-0001-8299-9361

Abstract

Dana Scott developed three presentations of the same class of domains: **continuous lattices** (1972), **neighbourhood systems** (PRG-19 / 1980–81), and **information systems** (1982). Each sibling formalization — [scott1972](#), [scott1980](#), [scott1982](#) — treats one presentation in isolation. This package (`ScottModels`) supplies the **bridge theorems**: order isomorphisms and constructivity audits showing that the three presentations determine the same domains (up to isomorphism), including products, separated sums, and function spaces at the information-system level, with a transport of 1972 Scott maps along the round presentation.

The Lean development is sorry-free. The continuous-lattice identification $D \simeq \text{RoundFilter}$ audits to `{propext, Quot.sound}`. Every `InfoSys`-facing declaration inherits `Classical.choice` from Mathlib 4.33’s `Finset` instances, even when the proof term extracts no witness. Genuine classical frontiers remain Scott’s 1972 topological definition of $\ll$ (algebraic $\Rightarrow$ continuous; `ScottMap` conjugation) and trichotomy for separated sums.

1 Introduction

Scott notes in the 1982 ICALP paper that neighbourhood systems and information systems are equivalent “in a precise sense.” The mathematical folklore is stronger still: all three presentations carve out the same class of domains, related by ideal completion and the Scott topology. That equivalence is already known; this article is the write-up of a machine-checked proof of the named bridges, not a first presentation of the result. The three presentations were formalized in separate libraries; the maps between them are the content of this package.

We do **not** re-prove Scott’s internal theorems; we import the finished sibling packages and build cross-presentation maps:

Presentation	Lean package	Characteristic object
Continuous lattices [Sco72]	<code>scott1972</code>	<code>IsContinuousLattice D</code> , way-below $\ll$, <code>ScottMap</code>
Neighbourhood systems [Sco81]	<code>scott1980</code>	<code>NeighborhoodSystem</code> , filters as domain elements
Information systems [Sco82]	<code>scott1982</code>	<code>InfoSys</code> , elements as consistent closed token sets

The main subtlety is that a naïve reading of “domains = filters of neighbourhoods” overstates the continuous-lattice corner: for a continuous lattice D , the filter $\{\uparrow a \mid a \leq x\}$ has retract x but properly contains the principal round filter $\{\uparrow a \mid a \ll x\}$. The correct identification is $D \simeq \mathbf{o} \text{ RoundFilter}$, not $\text{raw } \mathbb{I}$.

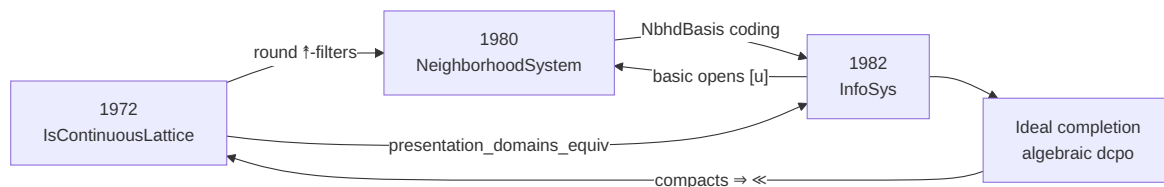


Figure 1: Introduction.

We prove **order isomorphisms of domains** (and of the named construction objects), not a 2-categorical equivalence of the full categories of continuous lattices / neighbourhood systems / information systems with all morphisms. Morphisms are linked where the sibling packages already supply them:

- approximable maps $\leftrightarrow$ Scott-continuous maps on $|A|$ (Factoid 4.6);
- Scott maps of continuous lattices $\leftrightarrow$ their conjugates on the round presentation.

A full functorial equivalence (preserving products, exponentials, and inverse limits simultaneously across all three presentations) would require additional coherence theorems.

2 Catalog of bridge theorems

Theorem	Direction	Lean module	Scope
<code>continuousLattice_to_neighborhoodSystem</code>	1972 $\rightarrow$ 1980	<code>ContinuousLatticeToNeighborhood.lean</code>	library
<code>neighborhoodSystem_to_infoSys</code>	1980 $\rightarrow$ 1982	<code>NeighborhoodToInfoSys.lean</code>	library
<code>infoSys_to_neighborhoodSystem</code>	1982 $\rightarrow$ 1980	<code>InfoSysToNeighborhood.lean</code>	library
<code>infoSys_to_idealCompletion</code>	1982 $\rightarrow$ algebraic	<code>InfoSysToIdealCompletion.lean</code>	library
<code>idealCompletion_to_continuousLattice</code>	algebraic $\rightarrow$ 1972	<code>IdealCompletionToContinuousLattice.lean</code>	library
<code>presentation_domains_equiv</code>	three-way	<code>PresentationDomains.lean</code>	library
<code>infoSys_constructions_equiv</code>	constructions	<code>InfoSysConstructions.lean</code> <code>ScottMapBridge.lean</code>	library

Theorem	Direction	Lean module	Scope
<code>sexNeighborhoodIso</code> / <code>sexIdealIso</code> / <code>sexDomainEquationIso</code>	S-expression instance	<code>WorkedExampleSEExpr.lean</code> <code>SexDomainEquation.lean</code>	compared

The sibling packages are **finished dependencies**, not work items of this paper: [scott1972](#), [scott1980](#), and [scott1982](#) (information systems through Factoid 8.4 / domain equations). They are vendored in `vendor/` at frozen SHAs (see `vendor/FROZEN.txt` / `lakefile.toml`), with the local modifications recorded there and in `formalization.yaml`; the remotes remain the per-paper homes. The compared family (Palomar `Challenge.lean` / `comparator.json`) is the 1980 $\leftrightarrow$ 1982 coding, the round three-way identification, and the Factoid 8.1 instance; the other catalog rows are proved in `ScottModels/` but are not Comparator targets. This package was verified and registered in the Palomar registry as [PALOMAR-2026-08-25-000003 v1](#) [[PALOMAR26](#)]. What Palomar verified is the claim:

Order isomorphisms relating Scott’s three presentations of domains — continuous lattices (1972), neighbourhood systems (PRG-19), and information systems (1982) — including the round identification $D \simeq \text{RoundFilter} \simeq \text{RoundInfoSysElement}$ and the S-expression equation $T \cong A + (T \times T)$ as $|T| \simeq |A + (T \times T)|$ (`SexSys.Element` $\simeq$ `SexRhs.Element`).

This document ([view.pdf](#)) is the narrative account, not a formalization of the three papers. The module map below is left-to-right: sibling packages on the left, local `ScottModels/` modules on the right, with arrows for **direct** imports (transitive imports inside the siblings are omitted). `Equivalence` re-exports the bridges; only the `ScottMapBridge` $\rightarrow$ `Equivalence` edge is drawn to avoid clutter. Root `ScottModels.lean` also opens 1972 `ContinuousLattice.Specialization`, 1980 `Neighborhood.Basic`, and 1982 `InfoSys`.

3 Proof notes

3.1 Continuous lattices $\rightarrow$ neighbourhood systems

Claim. For a continuous lattice D , the sets $\uparrow a = \{z \mid a \ll z\}$ form a `NeighborhoodSystem` on token type D , and under `IsContinuousLattice` one has an order isomorphism $D \simeq \text{RoundFilter}$.

Construction. `wayBelowUp` a is upward closed; $\uparrow \perp = \text{univ}$; $\uparrow a \cap \uparrow b = \uparrow(a \sqcup b)$ (interpolation / directedness of way-below). Principal filters `toFilter` $x = \{\uparrow a \mid a \ll x\}$ are round: membership of $\uparrow a$ yields, by interpolation in a continuous lattice, some b with $a \ll b \ll x$, hence $\uparrow b \in \text{toFilter } x$.

Why `raw` || fails. The larger filter $\{\uparrow a \mid a \leq x\}$ still has `offFilter` = x (supremum of codes), but is not equal to `toFilter` x whenever there are elements way-below strictly below x . Roundness ($\uparrow a \in f \Rightarrow \exists b, a \ll b \wedge \uparrow b \in f$) cuts exactly to the image of `toFilter`.

Axioms. `domainOrderIso` : $D \simeq \text{RoundFilter}$ audits to `{propext, Quot.sound}`.

3.2 Neighbourhood systems $\rightarrow$ information systems

Claim. A neighbourhood system equipped with a decidable exhaustive coding `NbhdBasis` ι α of its neighbourhood family induces an `InfoSys` on tokens ι with `||` $\simeq$ the `InfoSys` domain.

Construction. Tokens are neighbourhood indices. Consistency of a finite $u \subseteq \iota$ is membership of $\bigcap_{i \in u} \text{nbhd } i$ in (empty intersection = master set Δ). Entailment $u \vdash j$ means

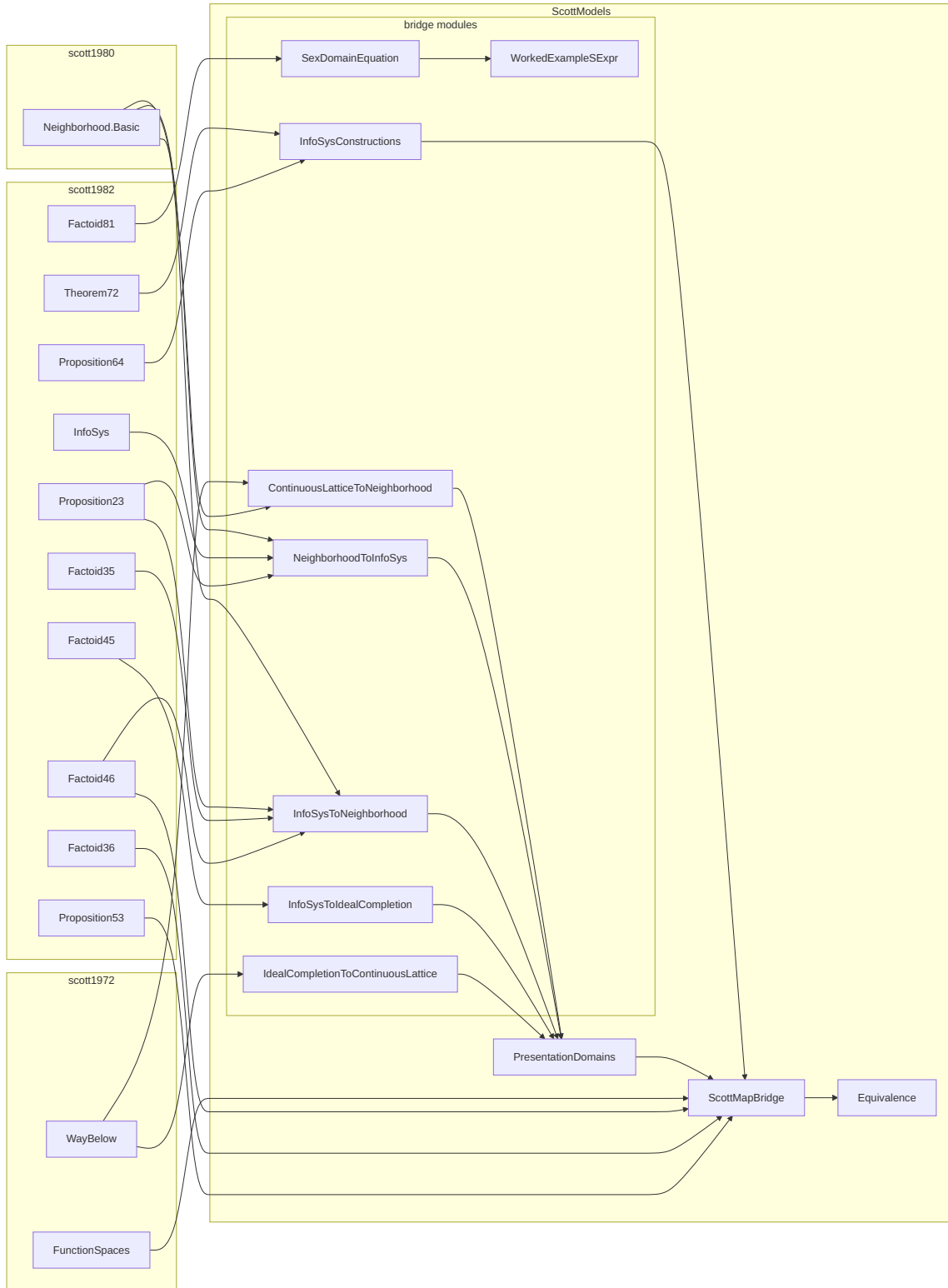


Figure 2: Catalog of bridge theorems.

the intersection is a neighbourhood contained in `nbhd j`. Filters of the neighbourhood system correspond to `InfoSys` elements via the coding.

Constructivity. `DecidableEq ι` is required so `InfoSys` can use the choice-free `Finset` prelude from `scott1982`. The proof of `ent_con` avoids classical `by_cases` / `em`. The maps extract no witnesses; the kernel audit still reports inherited `Classical.choice` from the Mathlib 4.33 `Finset` instance frontier (§4).

3.3 Information systems → neighbourhood systems

Claim. For an information system `A`, the basic opens $[u] = \{x \in |A| \mid \uparrow u \subseteq x\}$ ($u \in \text{Con}$) form a `NeighborhoodSystem` on `|A|`, and filters recover elements: $|A| \simeq_o$ the filter domain.

Proof note. Scott’s Factoid 4.6 supplies the basic-open vocabulary. The Lean proof initially pulled `Classical.choice` through `simp` on `basicOpen_empty` / `finset unions`; those `sims` were removed so the construction extracts no witnesses. The kernel audit still reports inherited `Classical.choice` through `InfoSys/Finset` (§4). Together with §3.2 this is the **1980** ↔ **1982** equivalence under coding.

3.4 Information systems → ideal completion

Claim. $|A| \simeq_o \text{Ideal}(\text{FiniteElement } A)$, where finite elements are closures of consistent finsets (Factoid 3.5).

Construction. `toIdeal x` is the ideal of finite approximants of `x`; `ofIdeal` takes directed suprema of finite elements (1982 Factoids 4.4–4.5: `directedSup`, `eq_directedSup`, `finiteApproximants`, `compact_closure`). The maps extract no witnesses; the kernel audit inherits `Classical.choice` through `InfoSys/Finset` (§4).

3.5 Algebraic complete lattices → continuous lattices

Claim. If every element of a complete lattice is the directed supremum of compact elements below it (`IsAlgebraicLattice`), then `IsContinuousLattice` holds.

Proof note. Order-theoretic compactness implies `Set.Ici k` is Scott-open, hence $k \ll y$ whenever $k \leq y$. Algebraicity then yields $y = \sqcup \{k \text{ compact} \mid k \leq y\} \subseteq \sqcup \{x \mid x \ll y\}$. This is the **classical frontier**: Scott’s $\ll$ is defined topologically in `scott1972`, so the footprint inherits that classicality even though the order argument is elementary.

3.6 Three-presentation equivalence (`presentation_domains_equiv`)

Claim. Under `IsContinuousLattice D` and `DecidableEq D`,

`D ?o RoundFilter ?o RoundInfoSysElement`

where `RoundInfoSysElement` is the subtype of elements of the `InfoSys` coded by `wayBelowNbhdBasis` (tokens = elements of `D`, neighbourhoods = $\uparrow a$) that correspond to round filters. An extended form routes through the ideal-completion subtype.

Glue. `wayBelowNbhdBasis` packages the $\uparrow$ -system as an `NbhdBasis`; `roundFilter_infoSys_iso` transports roundness along `NbhdBasis.domainOrderIso`; `presentation_domains_equiv` is the composite with `continuousLattice_roundFilter_iso`.

What is not claimed. `Raw ||` and the full `InfoSys` domain `|A|` remain properly larger than `D`. The equivalence is on the **round** corner that matches continuous lattice points.

Axioms. Under Mathlib 4.33 the kernel audit includes `{propext, Classical.choice, Quot.sound}` through the inherited `InfoSys/Finset` instance frontier described in §4; the round-filter construction itself performs no choice-based witness extraction.

3.7 Constructions (infoSys_constructions_equiv)

Products

$|A| \times |B| \simeq_o |A \times B|$ via `1982.pairElements / fstMap / sndMap` (Prop 6.2). The maps use explicit projections and assembly, with no witness selection.

Separated sums

$|A| \oplus |B| \simeq_o |A + B|$ via `inl / inr` classify-and-assemble (Prop 6.4). Trichotomy on token polarity uses classical case-split (`Classical.choice` in the footprint).

Function spaces

$\text{ApproximableMap } A \ B \simeq_o |A \rightarrow B|$ packages Theorem 7.2 (`approxMap_toElement / element_toApproxMap`) with pointwise `Le` as `PartialOrder`. No additional choice is used beyond the inherited `InfoSys/Finset` instance frontier.

Factoid 4.6 bridge

$\text{ApproximableMap } A \ B \simeq_o \text{ScottContinuous } A \ B$ via `toScottContinuous / ofScottContinuous`, using Prop 5.3(v) (`rel_iff_closure_le`) and `closure_le_element` for the round-trip on relations. The relation-level proof performs no witness extraction; its kernel audit inherits the `InfoSys/Finset` frontier in §4.

ScottMap conjugation

A 1972 `ScottMap` $D \ E$ conjugates along any pair of order isomorphisms $\iota_D : D \simeq_o D'$, $\iota_E : E \simeq_o E'$ to a pointwise-ordered map $D' \rightarrow E'$. Specializing to `continuousLattice_roundFilter_iso / presentation_domains_equiv` yields `scottMap_roundFilter_iso` and `scottMap_roundInfoSys_iso`. The conjugation itself is order-theoretic; the footprint includes `Classical.choice` because `ScottMap` is defined via Scott continuity in `scott1972`.

Out of scope (documented). Identifying `ApproximableMap` on the coded $\uparrow$ -`InfoSys` with `ScottMap` $D \ E$ (needs roundness preservation of approximable maps), and relating `wayBelow` ($D \times E$) to the `InfoSys` product of factors (cylinder basis).

4 Constructivity summary

Bridge	Footprint	Notes
$\text{Nbhd} \leftrightarrow \text{InfoSys}$ (both directions)	+ inherited <code>Classical.choice</code>	Decidable coding; Mathlib 4.33 <code>Finset</code> instance frontier
$\text{CL} \leftrightarrow \text{RoundFilter}$	<code>{propext, Quot.sound}</code>	Roundness is order-theoretic
$\text{InfoSys} \leftrightarrow \text{Ideal}$	+ inherited <code>Classical.choice</code>	Factoids 4.4–4.5; type-level <code>Finset</code> frontier
Algebraic $\Rightarrow$ CL	classical	1972 topological $\ll$
Product / function space (1982)	+ inherited <code>Classical.choice</code>	No witness extraction in the maps
Separated sum (1982)	+ <code>Classical.choice</code>	Token polarity trichotomy
Factoid 4.6 <code>ApproxMap</code> $\leftrightarrow$	+ inherited	Type-level <code>Finset</code> frontier
<code>ScottContinuous</code>	<code>Classical.choice</code>	
<code>ScottMap</code> conjugation	+ <code>Classical.choice</code>	Via 1972 <code>ScottMap</code>

Target discipline (from the 1982 package): prefer constructive proofs wherever Scott’s 1982 text emphasizes constructivity; call out classical frontiers explicitly rather than hiding choice in automation.

Mathlib 4.33 makes `Finset.instSetLike` and `Finset.instPartialOrder` depend on `Classical.choice`. Consequently `#print axioms` reports choice for declarations whose types traverse `InfoSys`,

even when their proof terms only manipulate explicit finite data. Here “no witness extraction” distinguishes that inherited library-instance footprint from genuine uses such as separated-sum polarity classification.

5 Worked example — S-expressions / trees

Lean packaging: [WorkedExampleSEExpr.lean](#), [SexDomainEquation.lean](#). The Palomar compared family includes `sexNeighborhoodIso`, `sexIdealIso`, and `sexDomainEquationIso` (Challenge / `comparator.json`). The compared domain-equation declaration is the order isomorphism `SexSys.Element` $\simeq_o$ `SexRhs.Element` ($|T| \simeq_o |A + (T \times T)|$), not the generic semantic factor `WithBot` ($|A| \oplus (|T| \times |T|) \simeq_o |A + (T \times T)|$).

5.1 Overview

The bridges of §§2–3 are abstract. This section fixes **one** concrete domain — Scott’s S-expression / tree equation $T \cong A + (T \times T)$ ([[Sco82](#)], Factoid 8.1) over the $\mathbb{N}$ lower-bound atom system (Factoid 2.4) — and asks what that domain looks like in each paradigm:

Paradigm	What an element of T is
1982	a consistent, deductively closed set of tree tokens
1980	a filter of basic opens <code>[u]</code> on that token domain
1972	a point of an algebraic continuous lattice (ideal of finite elements)

We build the domain natively as an information system (§5.2), then **read the same carrier** as a neighbourhood system (§5.3) and as a continuous lattice (§5.4). Section 5.5 exhibits the order isomorphisms for this instance — so that, on S-expressions,

$T_{1982} \simeq_o T_{1980} \simeq_o \text{Ideal}(K(T))$ and `Ideal(K(T))` is a continuous lattice,

hence transitively $T_{1982} \simeq_o T_{1980}$ and (via the algebraic $\Rightarrow$ continuous frontier) the 1972 presentation of the same domain. Order isomorphism is an equivalence relation, so the three presentations determine one domain up to $\simeq_o$. Section 5.6 records two extras that live on top of the carriers: the domain equation at the level of domains, and identity as a morphism in both the approximable-map and Scott-continuous languages.

5.2 Information system (1982)

Atoms are propositions $n \leq x$ on $\mathbb{N}$ (`lowerBoundSystem`). The tree system `SexSys := treeSystem lowerBoundSystem` has inductive tokens `TreeToken` $\mathbb{N}$ (`bot` / `atom` / `pairL` / `pairR`). Consistency and entailment are Scott’s sum $\times$ product clauses; Factoid 8.1 records that this is literally the information system of the right-hand side $A + (T \times T)$:

- `SexRhs = sumSystem A (productSystem T T) (sexRhs_eq_sum_product);`
- token unfolding `treeUnfold` sends `atom n` to the left summand (`sexUnfold_atom`) and is a retraction onto the official sum-of-product carrier, with section `treeFold`;
- `TreeEnt` matches official product `ent_bot` on the two encodings of (Δ, Δ) (`pairL` Δ and `pairR` Δ), so closed elements are saturated for `ker(treeUnfold)`.

Finite elements include singleton atom closures $\text{sexAtom } n = \text{for } u = \{\text{atom } n\}$. Elements of $|T|$ are the consistent closed sets of tokens — the 1982 presentation of the S-expression domain.

5.3 Neighbourhood system (1980)

From $|T|$ one obtains a neighbourhood system whose basic opens are $[u] = \{x \in |T| \mid \uparrow u \subseteq x\}$. In this paradigm an element of the domain is a **filter** of such opens (not a raw set of tokens). Nothing new is invented: it is the image of SexSys under $\text{InfoSysToNeighborhood}$. The concrete iso of §5.5 below is exactly “read each token-element as its filter of basic opens.”

5.4 Continuous lattice (1972)

Finite elements $K(T)$ are the closures of consistent finsets. In the algebraic style that feeds Scott’s 1972 theory, a domain element is an **ideal** of $K(T)$ — the ideal completion of the compact basis. That poset of ideals is an algebraic complete lattice, hence a continuous lattice ($\text{IsAlgebraicLattice} \Rightarrow \text{IsContinuousLattice}$, §3.5). This is the 1972 presentation of the same S-expression domain: points ordered by inclusion of ideals, with way-below recovered from the compact basis.

We do not claim a choice-free $D \simeq \text{RoundFilter}$ story built from trees alone; the round $\uparrow$ -filter identification of §3.1 applies once one is already in the continuous-lattice setting.

5.5 Equivalence for this instance

Specializing the bridges of §3 to SexSys yields order isomorphisms of carriers:

- $\text{sexNeighborhoodIso} : |T| \simeq \text{filters of } [u] \text{ (InfoSysToNeighborhood.domainOrderIso)}$
— 1982 $\simeq$ 1980;
- $\text{sexIdealIso} : |T| \simeq \text{Ideal (FiniteElement } T \text{) (InfoSysToIdealCompletion.domainOrderIso)}$
— 1982 $\simeq$ algebraic / 1972 carrier;
- $\text{sexNeighborhoodIdealIso (neighborhood_ideal_iso)}$ — the constructive triangle closing 1980 $\simeq$ $\text{Ideal}(K(T))$ without going back through tokens.

Composing any two edges recovers the third: isomorphism is transitive, so on this example the three presentations are pairwise equivalent. The 1972 corner uses the algebraic $\Rightarrow$ continuous implication already flagged in §3.5 / §4. The neighbourhood $\leftrightarrow$ information $\leftrightarrow$ ideal triangle inherits Classical.choice only through the Finset instance frontier of §4.

5.6 Domain equation and morphisms

Beyond identifying the carriers, the same example exercises constructions and maps.

Domain equation. Factoid 8.1’s token unfolding treeUnfold (section treeFold) lifts to an order isomorphism of the tree domain with the official RHS domain:

$\text{sexDomainEquationIso} : \text{SexSys.Element} \simeq \text{SexRhs.Element}$
($\text{treeDomainIso}; |T| \simeq |A| + (T \times T)|$). The maps are verified unfolding: the carrier of $\text{sexDomainEquationIso}$ x is $\text{treeUnfold } 'x.\text{carrier}$, and the carrier of the inverse is the preimage $\text{treeUnfold } ^{-1} y.\text{carrier}$ ($\text{sexDomainEquationIso_unfold} / \text{sexDomainEquationIso_fold}$). The only non-injectivity of treeUnfold is Scott’s two encodings of the product bottom ($\text{pairL } \Delta$ and $\text{pairR } \Delta$); after the ent_bot match, closed elements are saturated for that kernel, so image and preimage invert. The generic semantic factor $\text{WithBot } (|A| \oplus (|T| \times |T|)) \simeq \text{SexRhs.Element}$ is the uncomparing sexRhsSemanticIso (classical only through the 1982 sum trichotomy); it is **not** the compared fixed-point equation.

Morphisms. The identity approximable map `idMap T` (Prop 5.4) transports along `Factoid 4.6` to a Scott-continuous endomap `sexIdScottContinuous` with `toFun = id (sexId_toElement)` — the same endomap, named once in each morphism language.

Axioms. Object-level neighbourhood / ideal isos, `sexDomainEquationIso`, and `Factoid 4.6` for `idMap` audit to `{propext, Classical.choice, Quot.sound}` through the `InfoSys/Finset` frontier of §4. The uncomparing semantic factor `sexRhsSemanticIso` additionally uses the 1982 sum trichotomy.

6 Build

```
lake exe cache get
lake build ScottModels
```

Pinned: Lean / mathlib v4.33.0. Sibling libraries are compiled from in-tree `vendor/` (`srcDir`; frozen revisions and local patches in `vendor/FROZEN.txt`): `scott1972 278ba1745b6fb023fc3e9ffe53` `scott1980 ef6c05671582b7883ecd40b7081b642ce5351963`, `scott1982 d3221eec09505aa75e1b818aabc73f1f` Palomar compared family: `Challenge.lean`, `Solution.lean`, `comparator.json`. The narrative paper of record is this document ([view.pdf](#)).

Acknowledgments (Dana Scott, AI tool cards, artifact URL) are injected before References when building `arxiv.tex` via `scripts/ai_model_cards.py` — they are not kept in this file.

Rebuild this document (PDF with complete Lean source appendix, one subsection per module):

```
bash scripts/build_arxiv_pdf.sh          # expand Lean -> tex -> arxiv.pdf
bash scripts/build_arxiv_pdf.sh --pdf-only # PDF only when arxiv.tex already current
```

7 Acknowledgments

- **Dana Scott** — *Continuous Lattices* [Sco72], *Lectures on a Mathematical Theory of Computation* (PRG-19) [Sco81], and *Domains for Denotational Semantics* [Sco82], the three presentations this development relates.

7.1 AI-assisted development

The human author retains sole responsibility for the mathematical content, the choice of formalization route, and every formal claim in this work. Following standard publisher practice (e.g., COPE guidance on authorship and AI tools [COPE24]), **no large language model is listed as a co-author** — authorship implies an accountability that automated systems cannot bear.

Because this development may borrow Lean and proof patterns from the sibling formalizations `scott1972` [SR72], `scott1980` [ER80], and `scott1982` [SR82], **every model in the registry is treated as used** for acknowledgement purposes. We gratefully acknowledge assistance from the following tools (auto-generated from `scripts/ai_model_cards.py` when building `arxiv.tex`):

- **Cursor** [Cur26] — agent-assisted editing in the Cursor IDE: bridge theorems relating Scott’s 1972 / 1980 / 1982 presentations in Lean 4 / mathlib, `lake build` repair, drafting this narrative (`arxiv.md`), and tracking the formalized inventory. Generated Lean was provisional until it compiled under the pinned toolchain. Also used in the sibling formalizations whose portable work may be copied into this repo.

- **Cursor Composer 2.5 Fast [Cmp25]** — routine multi-step work: module scaffolding, dependency-ordered wiring of `ScottModels/`, documentation and Mermaid blueprints, and medium proof obligations where the strategy was already fixed (`composer-2.5`). Also used in sibling repos whose portable Lean may be adapted here.
- **Cursor Grok 4.5 [Grk45]** — primary agent model for the `scott_models` sessions: bridge theorems (`presentation_domains_equiv`, round filters, constructions), inventory design in `arxiv.md`, constructivity audits, and adaptation of portable patterns from prior Scott formalizations. Jointly trained by SpaceXAI and Cursor; used here via the Cursor agent environment.
- **Anthropic Claude Sonnet 5 (medium reasoning) [Son26]** — day-to-day formalization and proof-engineering in Cursor at the medium reasoning tier in sibling Scott formalizations (and available for this repo): inventory and narrative maintenance, module wiring, and medium-complexity Lean obligations where the proof strategy was already fixed. Listed because portable prior work may be copied into `scott_models`.
- **Anthropic Claude Opus 4.8 (high reasoning) [Ant26]** — selective use for the heaviest proof work in sibling formalizations (e.g. continuous lattices and PRG-19). Every emitted proof term was checked by the Lean kernel. Listed because portable prior work may be copied into `scott_models`.
- **Google Gemini 3.5 Flash [Gem25]** — exploratory passes in sibling formalizations (typographic conventions, scope decisions). Listed because portable prior work may be copied into `scott_models`.

All definitions, constructivity audits, and final prose were reviewed by the human author, who takes full responsibility for them.

7.2 Artifact availability and reproducibility

The development is at github.com/catskillsresearch/scott_models. GitHub Actions CI checks out the three sibling packages as path dependencies and runs `lake build`; the library builds successfully on `main` (green status).

Locally:

```
lake exe cache get
lake build ScottModels
```

Sibling packages `scott1972`, `scott1980`, `scott1982` are Lake path dependencies (`lakefile.toml`). Session state lives in `HANDOFF.md`; `arxiv.md` is the durable inventory and proof narrative. Axiom audits: `#print axioms` on the blueprint-facing names (`presentation_domains_equiv`, `infoSys_product_domain_equiv`, `approximableMap_scottContinuous_equiv`, `scottMap_roundInfoSys_iso`, ...).

Build the arXiv PDF / submission zip (Lean Code appendix = GitHub hyperlinks):

```
bash scripts/build_arxiv_tex.sh      # arxiv.md -> arxiv.tex + figures/
bash scripts/build_arxiv_pdf.sh      # compile PDF + package dist/arxiv_submit.zip
# or: bash scripts/package_arxiv_submit.sh
```

List of Figures

1	Introduction.	2
2	Catalog of bridge theorems.	4

8 References

- [Sco72] Dana Scott. *Continuous Lattices*. In F. W. Lawvere (ed.), *Toposes, Algebraic Geometry and Logic*, LNM 274, Springer, 1972.
- [Sco81] Dana Scott. *Lectures on a Mathematical Theory of Computation*. Technical Monograph PRG-19, Oxford University Computing Laboratory, 1981 (neighbourhood systems).
- [Sco82] Dana Scott. *Domains for Denotational Semantics*. ICALP 1982, LNCS 140, Springer, 1982.
- [AJ94] S. Abramsky and A. Jung. *Domain Theory*. In *Handbook of Logic in Computer Science*, Vol. 3, Oxford University Press, 1994.
- [GHKLMS03] G. Gierz et al. *Continuous Lattices and Domains*. Cambridge University Press, 2003.
- [SR72] Companion Lean formalization: [scott1972](#).
- [ER80] Companion Lean formalization: [scott1980](#).
- [SR82] Companion Lean formalization: [scott1982](#).
- [PALOMAR26] Palomar Registry. Entry PALOMAR-2026-08-25-000003, version 1. Verified claim: order isomorphisms relating Scott’s three presentations of domains — continuous lattices (1972), neighbourhood systems (PRG-19), and information systems (1982) — including the round identification $D \simeq_o \text{RoundFilter} \simeq_o \text{RoundInfoSysElement}$ and the S-expression equation $T \cong A + (T \times T)$ as $|T| \simeq_o |A + (T \times T)|$ ($\text{SexSys.Element} \simeq_o \text{SexRhs.Element}$). <https://palomar-registry.org/entry?id=PALOMAR-2026-08-25-000003&version=1>
- [COPE24] Committee on Publication Ethics (COPE). *Authorship and AI tools: COPE position statement*. 2024. <https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools>
- [Cur26] Anysphere, Inc. *Cursor: AI-native code editor and agent environment*. <https://cursor.com> (accessed 2026).
- [Cmp25] Anysphere, Inc. *Composer 2.5*. Model announcement and documentation, <https://cursor.com/blog/composer-2-5>; model card as integrated in Cursor, <https://cursor.com/docs/models> (accessed 2026).
- [Grk45] SpaceXAI and Anysphere, Inc. *Grok 4.5*. Model documentation, <https://docs.x.ai/developers/models/grok-4.5>; Cursor announcement, <https://cursor.com/blog/grok-4-5> (accessed 2026).
- [Son26] Anthropic. *Claude Sonnet 5* (medium reasoning variant). System card, <https://www.anthropic.com/claude-sonnet-5-system-card>; model documentation as integrated in Cursor, <https://cursor.com/docs/models> (accessed 2026).
- [Ant26] Anthropic. *Claude Opus 4.8* (high thinking/reasoning variant). System card and announcement, <https://www.anthropic.com/news/claude-opus-4-8>; model documentation as integrated in Cursor, <https://cursor.com/docs/models/claude-opus-4-8> (accessed 2026).

- [Gem25] Google DeepMind. *Gemini 3.5 Flash*. Technical documentation and model cards. <https://ai.google.dev/gemini-api/docs/models> (accessed 2026).

A Complete Lean source

Role	File	Lines
Root import graph	ScottModels.lean	21
Bridge	ScottModels/NeighborhoodToInfoSys.lean	267
Bridge	ScottModels/InfoSysToNeighborhood.lean	246
Bridge	ScottModels/ContinuousLatticeToNeighborhood.lean	228
Bridge	ScottModels/InfoSysToIdealCompletion.lean	183
Bridge	ScottModels/IdealCompletionToContinuousLattice.lean	78
Bridge / packaging	ScottModels/PresentationDomains.lean	178
Bridge / packaging	ScottModels/InfoSysConstructions.lean	495
Bridge / packaging	ScottModels/ScottMapBridge.lean	190
Bridge / packaging	ScottModels/SexDomainEquation.lean	644
Bridge / packaging	ScottModels/WorkedExampleSEExpr.lean	173
Bridge / packaging	ScottModels/Equivalence.lean	41

Total: 12 files, 2744 lines of Lean.

Files appear in ScottModels.lean import order. Each block is a verbatim copy of the repository file at generation time.

A.1 ScottModels.lean

21 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Scott1972.ContinuousLattice.Specialization
import Scott1980.Neighborhood.Basic
import Scott1982.InfoSys
import ScottModels.NeighborhoodToInfoSys
import ScottModels.InfoSysToNeighborhood
import ScottModels.ContinuousLatticeToNeighborhood
import ScottModels.InfoSysToIdealCompletion
import ScottModels.IdealCompletionToContinuousLattice
import ScottModels.PresentationDomains
import ScottModels.InfoSysConstructions
import ScottModels.ScottMapBridge
import ScottModels.SexDomainEquation
import ScottModels.WorkedExampleSEExpr
import ScottModels.Equivalence

```

A.2 ScottModels/NeighborhoodToInfoSys.lean

267 lines.

Lean 4 source

```
/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott_models
-/

import Mathlib.Data.Finset.Fold
import Mathlib.Order.Hom.Basic
import Scott1980.Neighborhood.Basic
import Scott1982.InfoSys
import Scott1982.Proposition23

/-!
# Neighbourhood systems -> information systems (1980 -> 1982)

Scott (1982) notes that neighbourhood systems and information systems are equivalent
in a precise sense. This module realises one direction: given a neighbourhood system
presented with a *decidable index of its neighbourhoods* (a constructive coding of
`?`), build an information system on those indices whose domain is order-isomorphic
to the original filter domain.

Tokens of the information system are indices of neighbourhoods; consistency is
membership of the finite intersection in `?` (Scott's empty intersection = `Delta`);
entailment is ?the intersection is a neighbourhood and is contained in the target.?
-/

namespace ScottModels

open Scott1980.Neighborhood
open Scott1982

/-- A neighbourhood system with a decidable exhaustive index of its neighbourhoods.
`DecidableEq ?` is required so the induced information system can use choice-free
`Finset` operations (`InfoSys` tokens). -/
structure NbhdBasis (? a : Type*) [DecidableEq ?] where
  system : NeighborhoodSystem a
  nbhd : ? -> Set a
  nbhd_mem : forall i, system.mem (nbhd i)
  exhaustive : forall {X : Set a}, system.mem X -> exists i, nbhd i = X
  botIdx : ?
  botIdx_eq : nbhd botIdx = system.master

namespace NbhdBasis

variable {? a : Type*} [DecidableEq ?] (B : NbhdBasis ? a)

/-- Finite intersection of coded neighbourhoods, with Scott's convention `?{} = Delta`. -/
def interOf (u : Finset ?) : Set a :=
  u.fold (* I *) B.system.master B.nbhd

@[simp] theorem interOf_empty : B.interOf ({} : Finset ?) = B.system.master := rfl

theorem interOf_insert {a : ?} {s : Finset ?} (ha : a notin s) :
  B.interOf (insert a s) = B.nbhd a I B.interOf s :=
  Finset.fold_insert (op := (* I *)) ha

theorem nbhd_subset_master (i : ?) : B.nbhd i subseq B.system.master :=
```

```

B.system.sub_master (B.nbhd_mem i)

theorem interOf_singleton (a : ?) : B.interOf ({a} : Finset ?) = B.nbhd a := by
  simp [interOf, Finset.fold_singleton, Set.inter_eq_left.mpr (B.nbhd_subset_master a)]

/-- Membership in the folded intersection. -/
theorem mem_interOf {u : Finset ?} {x : a} :
  x in B.interOf u <-> x in B.system.master /\ forall i in u, x in B.nbhd i :=
  by
  induction u using Finset.induction with
  | empty =>
    simp [interOf_empty]
  | insert a s ha ih =>
    rw [B.interOf_insert ha, Set.mem_inter_iff, ih]
    constructor
  * rintro <ha', hm, hs>
    refine <hm, fun i hi => ?_>
    rcases Finset.mem_insert.mp hi with rfl | hi
    * exact ha'
    * exact hs i hi
  * rintro <hm, hall>
    exact <hall a (Finset.mem_insert_self a s), hm,
      fun i hi => hall i (Finset.mem_insert_of_mem hi)>

theorem interOf_subset_nbhd {u : Finset ?} {i : ?} (hi : i in u) :
  B.interOf u subseq B.nbhd i := fun _ hx => (B.mem_interOf.mp hx).2 i hi

theorem interOf_subset_master (u : Finset ?) : B.interOf u subseq B.system.master :=
  fun _ hx => (B.mem_interOf.mp hx).1

/-- Larger index sets give smaller intersections. -/
theorem interOf_anti {u v : Finset ?} (h : u subseq v) : B.interOf v subseq B.interOf u
:= by
  intro x hx
  exact B.mem_interOf.mpr <(B.mem_interOf.mp hx).1, fun i hi => (B.mem_interOf.mp hx).2 i (h hi)>

/-- If a neighbourhood sits below `interOf u`, then `interOf u` in `?`. -/
theorem interOf_mem_of_lower_bound {u : Finset ?} {Z : Set a}
  (hZ : B.system.mem Z) (hsub : Z subseq B.interOf u) : B.system.mem (B.interOf u) := by
  induction u using Finset.induction with
  | empty =>
    simp [interOf_empty] using B.system.master_mem
  | insert a s ha ih =>
    rw [B.interOf_insert ha] at hsub | -
    have hs : B.system.mem (B.interOf s) := ih (hsub.trans Set.inter_subset_right)
    exact B.system.inter_mem (B.nbhd_mem a) hs hZ hsub

/-- A filter containing every `nbhd i` for `i in u` contains `interOf u`. -/
theorem filter_mem_interOf (x : B.system.Element) {u : Finset ?}
  (hu : forall i in u, x.mem (B.nbhd i)) : x.mem (B.interOf u) := by
  induction u using Finset.induction with
  | empty =>
    simp [interOf_empty] using x.master_mem
  | insert a s ha ih =>
    rw [B.interOf_insert ha]
    exact x.inter_mem (hu a (Finset.mem_insert_self a s))
      (ih fun i hi => hu i (Finset.mem_insert_of_mem hi))

/-- **neighborhoodSystem_to_infoSys** -/
def toInfoSys : InfoSys ? where
  bot := B.botIdx
  Con := {u | B.system.mem (B.interOf u)}

```

```

Ent := fun u a => B.system.mem (B.interOf u) /\ B.interOf u subseq B.nbhd a
con_subset := by
  intro u v hu hv
  exact B.interOf_mem_of_lower_bound hu (B.interOf_anti hv)
con_sing := by
  intro a
  simp [Set.mem_setOf_eq, B.interOf_singleton] using B.nbhd_mem a
ent_con := by
  intro u a <hu, hsub>
  have h_1 : B.interOf (insert a u) subseq B.interOf u :=
    B.interOf_anti (Finset.subset_insert a u)
  have h_2 : B.interOf u subseq B.interOf (insert a u) := by
    intro x hx
    refine B.mem_interOf.mpr <(B.mem_interOf.mp hx).1, fun i hi => ?_>
    rcases Finset.mem_insert.mp hi with rfl | hi
    * exact hsub hx
    * exact (B.mem_interOf.mp hx).2 i hi
  have heq : B.interOf (insert a u) = B.interOf u := Set.Subset.antisymm h_1 h_2
  simp [Set.mem_setOf_eq, heq] using hu
ent_bot := by
  intro u hu
  refine <hu, ?_>
  intro x hx
  rw [B.botIdx_eq]
  exact B.interOf_subset_master u hx
ent_refl := by
  intro u a hu ha
  exact <hu, B.interOf_subset_nbhd ha>
ent_trans := by
  intro u v c hv _hu hall <_, hsub>
  refine <hv, ?_>
  intro x hx
  have hxu : x in B.interOf u :=
    B.mem_interOf.mpr <B.interOf_subset_master v hx, fun i hi => (hall i hi).2 hx>
  exact hsub hxu

/-! ## Domain isomorphism -/

/-- Filter -> InfoSys element. -/
def toElement (x : B.system.Element) : B.toInfoSys.Element where
  carrier := {i | x.mem (B.nbhd i)}
  consistent := fun Y hY => x.sub (B.filter_mem_interOf x hY)
  closed := by
    intro Y a hY <_, hsub>
    exact x.up_mem (B.filter_mem_interOf x hY) (B.nbhd_mem a) hsub

/-- InfoSys element -> filter (upward closure of its coded neighbourhoods). -/
def ofElement (e : B.toInfoSys.Element) : B.system.Element where
  mem X := B.system.mem X /\ exists i in e.carrier, B.nbhd i subseq X
  sub h := h.1
  master_mem := by
    refine <B.system.master_mem, B.botIdx, ?_, by rw [B.botIdx_eq]>
    have hEnt : B.toInfoSys.Ent {} B.botIdx := B.toInfoSys.ent_bot (InfoSys.con_empty _)
    exact e.closed {} B.botIdx (fun _ h => False.elim (Finset.notMem_empty _ h)) hEnt
  inter_mem := by
    intro X Y <hX, i, hi, hix> <hY, j, hj, hjy>
    have hCon : ({i, j} : Finset ?) in B.toInfoSys.Con :=
      e.consistent {i, j} (by
        intro x hx
        rcases Finset.mem_insert.mp hx with rfl | hx
        * exact hi
        * have : x = j := Finset.mem_singleton.mp hx
          exact this |> hj)

```

```

have hInterMem : B.system.mem (B.interOf ({i, j} : Finset ?)) := hCon
have hInterSub : B.interOf ({i, j} : Finset ?) subseq X I Y := by
  intro x hx
  refine <hix (B.interOf_subset_nbhd (by simp) hx),
    hji (B.interOf_subset_nbhd (by simp) hx)>
refine <B.system.inter_mem hX hY hInterMem hInterSub, ?_>
obtain <k, hk> := B.exhaustive hInterMem
refine <k, ?_, ?_>
* have hEnt : B.toInfoSys.Ent {i, j} k := <hInterMem, fun x hx => hk |> hx>
  exact e.closed {i, j} k (by
    intro x hx
    rcases Finset.mem_insert.mp hx with rfl | hx
    * exact hi
    * exact Finset.mem_singleton.mp hx |> hj) hEnt
* intro x hx
  exact hInterSub (hk |> hx)
up_mem := by
  intro X Y <hX, i, hi, hix> hY hXY
  exact <hY, i, hi, hix.trans hXY>

theorem toElement_carrier (x : B.system.Element) :
  (B.toElement x).carrier = {i | x.mem (B.nbhd i)} := rfl

theorem ofElement_mem (e : B.toInfoSys.Element) (X : Set a) :
  (B.ofElement e).mem X <-> B.system.mem X /\ exists i in e.carrier, B.nbhd i
  subseq X :=
  Iff.rfl

theorem toElement_ofElement (e : B.toInfoSys.Element) :
  B.toElement (B.ofElement e) = e := by
  have hc : (B.toElement (B.ofElement e)).carrier = e.carrier := by
    ext i
    constructor
  * intro hi
    rcases (B.ofElement_mem e (B.nbhd i)).mp hi with <_, j, hj, hsub>
    have hEnt : B.toInfoSys.Ent {j} i := by
      refine <by simp [B.interOf_singleton] using B.nbhd_mem j, ?_>
      simp [B.interOf_singleton] using hsub
    exact e.closed {j} i (by
      intro x hx
      have : x = j := Finset.mem_singleton.mp hx
      exact this |> hj) hEnt
  * intro hi
    exact <B.nbhd_mem i, i, hi, subset_rfl>
rcases e with <c, cons, clo>
rcases hte : B.toElement (B.ofElement <c, cons, clo>) with <c', cons', clo'>
have hc' : c' = c := by
  simp [hte, toElement_carrier] using hc
subst hc'
rfl

theorem ofElement_toElement (x : B.system.Element) :
  B.ofElement (B.toElement x) = x := by
  refine NeighborhoodSystem.Element.ext (V := B.system) fun X => ?_
  constructor
  * intro <hX, i, hi, hsub>
    exact x.up_mem hi hX hsub
  * intro hX
    obtain <i, rfl> := B.exhaustive (x.sub hX)
    exact <x.sub hX, i, hX, subset_rfl>

/-- Order isomorphism between the neighbourhood-system domain and the induced
information-system domain. -/

```

```

def domainOrderIso : B.system.Element ?o B.toInfoSys.Element where
  toFun := B.toElement
  invFun := B.ofElement
  left_inv := B.ofElement_toElement
  right_inv := B.toElement_ofElement
  map_rel_iff' := by
    intro x y
    constructor
    * intro h X hX
      obtain <i, rfl> := B.exhaustive (x.sub hX)
      have hi : i in (B.toElement x).carrier := hX
      have hi' : i in (B.toElement y).carrier := h hi
      exact y.up_mem hi' (x.sub hX) subset_rfl
    * intro h i hi
      exact h (B.nbhd i) hi

end NbhdBasis

/-- Blueprint name: neighbourhood system (with decidable basis) -> information system. -/
abbrev neighborhoodSystem_to_infoSys {? a : Type*} [DecidableEq ?] (B : NbhdBasis ? a) :
  InfoSys ? :=
  B.toInfoSys

end ScottModels

```

A.3 ScottModels/InfoSysToNeighborhood.lean

246 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Scott1980.Neighborhood.Basic
import Scott1982.Factoid35
import Scott1982.Factoid46
import Scott1982.Proposition23

/-!
# Information systems -> neighbourhood systems (1982 -> 1980)

Scott (1982, S4): for `u in Con`, the basic open
`[u] = { x in |A| | u subseq x }` yields a neighbourhood system on the domain `|A|`
whose filters recover the original elements. This is the converse direction to
`NeighborhoodToInfoSys`.
-/

namespace ScottModels

open Scott1980.Neighborhood
open Scott1982
open Scott1982.Constructive
open Scott1982.InfoSys

namespace InfoSysToNeighborhood

```

```

variable {a : Type*} [DecidableEq a] (A : InfoSys a)

theorem basicOpen_empty : A.basicOpen ({ } : Finset a) = (Set.univ : Set A.Element) := by
  ext x
  constructor
  * intro; trivial
  * intro _ a ha
    exact False.elim (Finset.notMem_empty a (Finset.mem_coe.1 ha))

theorem mem_basicOpen_singleton {a : a} {x : A.Element} :
  x in A.basicOpen ({a} : Finset a) <-> a in x.carrier := by
  constructor
  * intro h
    exact h (Finset.mem_coe.2 (Finset.mem_singleton_self a))
  * intro ha b hb
    have hb' : b = a := Finset.mem_singleton.mp (Finset.mem_coe.1 hb)
    exact hb' |> ha

theorem funion_singleton_eq_insert (a : a) (s : Finset a) :
  ({a} : Finset a) ∪ s = insert a s := by
  ext x
  constructor
  * intro hx
    rcases mem_funion.mp hx with h | h
    * exact Finset.mem_insert.mpr (Or.inl (Finset.mem_singleton.mp h))
    * exact Finset.mem_insert_of_mem h
  * intro hx
    rcases Finset.mem_insert.mp hx with h | h
    * exact mem_funion.mpr (Or.inl (h |> Finset.mem_singleton_self a))
    * exact mem_funion.mpr (Or.inr h)

theorem basicOpen_singleton_inter (a : a) (s : Finset a) :
  A.basicOpen ({a} : Finset a) ∩ A.basicOpen s = A.basicOpen (insert a s) := by
  rw [basicOpen_inter A, funion_singleton_eq_insert]

theorem ent_of_basicOpen_subset {u w : Finset a} (hw : w in A.Con)
  (hsub : A.basicOpen w ⊆ A.basicOpen u) {a : a} (ha : a in u) : A.Ent w a := by
  have : A.closure w hw in A.basicOpen w := subset_closure A hw
  have hU : A.closure w hw in A.basicOpen u := hsub this
  exact hU (Finset.mem_coe.2 ha)

theorem con_of_basicOpen_subset {u w : Finset a} (hw : w in A.Con)
  (hsub : A.basicOpen w ⊆ A.basicOpen u) : u in A.Con := by
  have hEnt : A.EntSet w u := fun a ha => ent_of_basicOpen_subset A hw hsub ha
  exact A.con_subset (proposition_2_3_ii A hw hEnt) (subset_funion_right _ _)

/-- **`infoSys_to_neighborhoodSystem`** -/
def toNeighborhoodSystem : NeighborhoodSystem A.Element where
  mem X := exists u, u in A.Con /\ X = A.basicOpen u
  master := Set.univ
  master_nonempty := <A.closure {} A.con_empty, Set.mem_univ _>
  master_mem := <{}, A.con_empty, (basicOpen_empty A).symm>
  inter_mem := by
    intro X Y Z hX hY hZ hZsub
    obtain <u, hu, rfl> := hX
    obtain <v, hv, rfl> := hY
    obtain <w, hw, rfl> := hZ
    have hsub : A.basicOpen w ⊆ A.basicOpen (u ∪ v) := by
      intro x hx
      have hx' : x in A.basicOpen u ∩ A.basicOpen v := hZsub hx
      rwa [basicOpen_inter A] at hx'
    have huv : u ∪ v in A.Con := con_of_basicOpen_subset A hw hsub
    exact <u ∪ v, huv, basicOpen_inter A u v>

```

```

sub_master := fun {_} _ => Set.subset_univ _

theorem mem_basicOpen_of_singletons (f : (toNeighborhoodSystem A).Element) {Y : Finset a}
  (hY : forall a in Y, f.mem (A.basicOpen ({a} : Finset a))) :
  f.mem (A.basicOpen Y) := by
  induction Y using Finset.induction with
  | empty =>
    rw [basicOpen_empty]
    have hmaster : (toNeighborhoodSystem A).master = (Set.univ : Set A.Element) := rfl
    rw [← hmaster]
    exact f.master_mem
  | insert a s _ha ih =>
    have hA := hY a (Finset.mem_insert_self a s)
    have hSm := ih fun i hi => hY i (Finset.mem_insert_of_mem hi)
    simpa [basicOpen_singleton_inter A a s] using f.inter_mem hA hSm

def toFilter (x : A.Element) : (toNeighborhoodSystem A).Element where
  mem U := exists u, u in A.Con /\ U = A.basicOpen u /\ u subseq x.carrier
  sub := by
    intro U h
    obtain <u, hu, rfl, _> := h
    exact <u, hu, rfl>
  master_mem := <{}, A.con_empty, (basicOpen_empty A).symm, by
    intro a ha
    exact False.elim (Finset.notMem_empty a (Finset.mem_coe.1 ha))>
  inter_mem := by
    intro U V hU hV
    obtain <u, hu, rfl, huX> := hU
    obtain <v, hv, rfl, hvX> := hV
    refine <u U ' v, ?_, basicOpen_inter A u v, ?_>
    * have hsub : (u U ' v) subseq x.carrier := by
      intro a ha
      rcases mem_funion.mp (Finset.mem_coe.1 ha) with h | h
      * exact huX (Finset.mem_coe.2 h)
      * exact hvX (Finset.mem_coe.2 h)
      exact x.consistent (u U ' v) hsub
    * intro a ha
      rcases mem_funion.mp (Finset.mem_coe.1 ha) with h | h
      * exact huX (Finset.mem_coe.2 h)
      * exact hvX (Finset.mem_coe.2 h)
  up_mem := by
    intro U V hU hV hUV
    obtain <u, hu, rfl, huX> := hU
    obtain <v, hv, rfl> := hV
    refine <v, hv, rfl, ?_>
    intro a ha
    exact x.closed u a huX (ent_of_basicOpen_subset A hu hUV ha)

def ofFilter (f : (toNeighborhoodSystem A).Element) : A.Element where
  carrier := {a | f.mem (A.basicOpen ({a} : Finset a))}
  consistent := by
    intro Y hY
    have hmem : f.mem (A.basicOpen Y) :=
      mem_basicOpen_of_singletons A f fun a ha => hY (Finset.mem_coe.2 ha)
    obtain <u, hu, heq> := f.sub hmem
    have hsub : A.basicOpen u subseq A.basicOpen Y := by
      intro x hx; exact heq |> hx
    exact con_of_basicOpen_subset A hu hsub
  closed := by
    intro Y b hY hEnt
    have hmemY : f.mem (A.basicOpen Y) :=
      mem_basicOpen_of_singletons A f fun c hc => hY (Finset.mem_coe.2 hc)
    have hsub : A.basicOpen Y subseq A.basicOpen ({b} : Finset a) := by

```

```

    intro x hx c hc
    have eqcb : c = b := Finset.mem_singleton.mp (Finset.mem_coe.1 hc)
    rw [eqcb]
    exact x.closed Y b hx hEnt
    exact f.up_mem hmemY <({b} : Finset a), A.con_sing b, rfl> hsub

theorem mem_basicOpen_iff (f : (toNeighborhoodSystem A).Element) {u : Finset a}
  (_hu : u in A.Con) :
  f.mem (A.basicOpen u) <-> u subteq (ofFilter A f).carrier := by
  constructor
  * intro h a ha
    have hsub : A.basicOpen u subteq A.basicOpen ({a} : Finset a) := by
      intro x hx c hc
      have hc' : c = a := Finset.mem_singleton.mp (Finset.mem_coe.1 hc)
      subst hc'
      exact hx (Finset.mem_coe.2 ha)
    exact f.up_mem h <({a} : Finset a), A.con_sing a, rfl> hsub
  * intro hsub
    exact mem_basicOpen_of_singletons A f fun a ha => hsub (Finset.mem_coe.2 ha)

theorem toFilter_ofFilter (f : (toNeighborhoodSystem A).Element) :
  toFilter A (ofFilter A f) = f := by
  refine NeighborhoodSystem.Element.ext (V := toNeighborhoodSystem A) fun U => ?_
  constructor
  * intro h
    obtain <u, hu, hU, hsub> := h
    subst hU
    exact (mem_basicOpen_iff A f hu).mpr hsub
  * intro hU
    obtain <u, hu, heq> := f.sub hU
    subst heq
    exact <u, hu, rfl, (mem_basicOpen_iff A f hu).mp hU>

theorem ofFilter_toFilter (x : A.Element) : ofFilter A (toFilter A x) = x := by
  have hc : (ofFilter A (toFilter A x)).carrier = x.carrier := by
    ext a
    constructor
  * intro ha
    obtain <u, hu, heq, hsub> := ha
    have huA : A.closure u hu in A.basicOpen u := subset_closure A hu
    have hsing : A.closure u hu in A.basicOpen ({a} : Finset a) := by
      rw [heq]; exact huA
    have ha' : a in (A.closure u hu).carrier := (mem_basicOpen_singleton A).1 hsing
    exact x.closed u a hsub ha'
  * intro ha
    refine <({a} : Finset a), A.con_sing a, rfl, ?_>
    intro c hc
    have hc' : c = a := Finset.mem_singleton.mp (Finset.mem_coe.1 hc)
    exact hc' |> ha
  rcases x with <c, cons, clo>
  rcases hte : ofFilter A (toFilter A <c, cons, clo>) with <c', cons', clo'>
  have hc' : c' = c := by simpa [hte] using hc
  subst hc'
  rfl

def domainOrderIso : A.Element ?o (toNeighborhoodSystem A).Element where
  toFun := toFilter A
  invFun := ofFilter A
  left_inv := ofFilter_toFilter A
  right_inv := toFilter_ofFilter A
  map_rel_iff' := by
    intro x y
    constructor

```

```

* intro h a ha
  have hx : (toFilter A x).mem (A.basicOpen ({a} : Finset a)) :=
    <({a} : Finset a), A.con_sing a, rfl, by
      intro c hc
      have hc' : c = a := Finset.mem_singleton.mp (Finset.mem_coe.1 hc)
      exact hc' |> ha>
  have hy : (toFilter A y).mem (A.basicOpen ({a} : Finset a)) := h _ hx
  obtain <u, hu, heq, hsub> := hy
  have huA : A.closure u hu in A.basicOpen u := subset_closure A hu
  have hsing : A.closure u hu in A.basicOpen ({a} : Finset a) := by
    rw [heq]; exact huA
  have ha' : a in (A.closure u hu).carrier := (mem_basicOpen_singleton A).1 hsing
  exact y.closed u a hsub ha'
* intro h U hU
  obtain <u, hu, hUeq, huX> := hU
  subst hUeq
  exact <u, hu, rfl, fun a ha => h (huX ha)>

end InfoSysToNeighborhood

/-- Blueprint-facing name for the 1982 -> 1980 basic-open neighbourhood system. -/
abbrev infoSys_to_neighborhoodSystem {a : Type*} [DecidableEq a] (A : InfoSys a) :
  NeighborhoodSystem A.Element :=
  InfoSysToNeighborhood.toNeighborhoodSystem A

end ScottModels

```

A.4 ScottModels/ContinuousLatticeToNeighborhood.lean

228 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Scott1972.ContinuousLattice.WayBelow
import Scott1980.Neighborhood.Basic

/-!
# Continuous lattices -> neighbourhood systems (1972 -> 1980)

For a continuous lattice `D`, Scott's way-below neighbourhoods
`Up a = { z | a << z }` form a `NeighborhoodSystem` on token set `D`.

Arbitrary filters need not be principal (`{ Up a | a <= x }` is a filter with
`ofFilter = x` but properly contains `toFilter x`). The correct domain is the
**round** filters: `Up a in f` implies `exists b, a << b /\ Up b in f`. Under
`IsContinuousLattice`, `D ?o` round filters via `toFilter` / `ofFilter`.
-/

namespace ScottModels

open Scott1972.ContinuousLattice
open Scott1980.Neighborhood
open scoped Scott1972.ContinuousLattice

```

```

namespace ContinuousLatticeToNeighborhood

variable {D : Type*} [CompleteLattice D]

/-- `Up a = { z | a <= z }`. -/
def wayBelowUp (a : D) : Set D := {z | a <= z}

theorem wayBelowUp_bot : wayBelowUp (False : D) = (Set.univ : Set D) := by
  ext z
  exact <fun _ => trivial, fun _ => bot_wayBelow z>

theorem wayBelowUp_inter (a b : D) :
  wayBelowUp a ∩ wayBelowUp b = wayBelowUp (a sqsup b) := by
  ext z
  constructor
  * exact fun <ha, hb> => WayBelow.sup ha hb
  * intro h
    exact <WayBelow.le_trans le_sup_left h, WayBelow.le_trans le_sup_right h>

theorem wayBelowUp_anti {a b : D} (hab : a <= b) :
  wayBelowUp b ⊆ wayBelowUp a :=
  fun _ h => WayBelow.le_trans hab h

/-- Neighbourhood system of all `Up a`. -/
def toNeighborhoodSystem : NeighborhoodSystem D where
  mem X := exists a : D, X = wayBelowUp a
  master := Set.univ
  master_nonempty := <False, Set.mem_univ _>
  master_mem := <False, wayBelowUp_bot.symm>
  inter_mem := by
    intro X Y _Z hX hY _hZ _hZsub
    obtain <a, rfl> := hX
    obtain <b, rfl> := hY
    exact <a sqsup b, wayBelowUp_inter a b>
  sub_master := fun { _ } _ => Set.subset_univ _

/-- Filter of the `Up`-neighbourhood system on `D`. -/
abbrev Filter : Type _ :=
  (toNeighborhoodSystem : NeighborhoodSystem D).Element

/-- Principal filter of `Up`-neighbourhoods at `x`. -/
def toFilter (x : D) : Filter (D := D) where
  mem U := exists a : D, U = wayBelowUp a /\ a <= x
  sub := by
    intro U h
    obtain <a, <rfl, _>> := h
    exact <a, rfl>
  master_mem := <False, wayBelowUp_bot.symm, bot_wayBelow x>
  inter_mem := by
    intro U V hU hV
    obtain <a, <rfl, ha>> := hU
    obtain <b, <rfl, hb>> := hV
    exact <a sqsup b, <wayBelowUp_inter a b, WayBelow.sup ha hb>>
  up_mem := by
    intro U V hU hV hUV
    obtain <a, <rfl, ha>> := hU
    obtain <b, rfl> := hV
    exact <b, <rfl, hUV ha>>

theorem toFilter_mono {x y : D} (hxy : x <= y) : toFilter x <= toFilter y := by
  intro U hU
  obtain <a, <rfl, ha>> := hU
  exact <a, <rfl, ha.trans_le hxy>>

```

```

theorem mem_wayBelowUp_toFilter {x a : D} :
  (toFilter x).mem (wayBelowUp a) <-> a << x := by
  constructor
  * intro h
    obtain <b, <heq, hb>> := h
    have : x in wayBelowUp b := hb
    rwa [ <- heq ] at this
  * intro ha
    exact <a, <rfl, ha>>

/-- Roundness: `Up a in f` is witnessed by a finer code `b` with `a << b`. -/
def IsRound (f : Filter (D := D)) : Prop :=
  forall {a : D}, f.mem (wayBelowUp a) -> exists b : D, a << b /\ f.mem (wayBelowUp b)

/-- Codes whose `Up`-neighbourhoods lie in the filter. -/
def codes (f : Filter (D := D)) : Set D :=
  {a : D | f.mem (wayBelowUp a)}

theorem mem_codes_iff {f : Filter (D := D)} {a : D} :
  a in codes f <-> f.mem (wayBelowUp a) :=
  Iff.rfl

theorem bot_mem_codes (f : Filter (D := D)) : (False : D) in codes f := by
  have : f.mem (wayBelowUp (False : D)) := by
    rw [wayBelowUp_bot]
    exact f.master_mem
  exact this

theorem codes_nonempty (f : Filter (D := D)) : (codes f).Nonempty :=
  <False, bot_mem_codes f>

theorem codes_directed (f : Filter (D := D)) : DirectedOn (* <= *) (codes f) := by
  intro a ha b hb
  refine <a sqsup b, ?_, le_sup_left, le_sup_right>
  have : f.mem (wayBelowUp a I wayBelowUp b) := f.inter_mem ha hb
  rwa [wayBelowUp_inter] at this

theorem codes_lower (f : Filter (D := D)) {a b : D} (hba : b <= a)
  (ha : a in codes f) : b in codes f :=
  f.up_mem ha <b, rfl> (wayBelowUp_anti hba)

/-- Retraction: `sqsup` of codes present in the filter. -/
noncomputable def ofFilter (f : Filter (D := D)) : D :=
  sSup (codes f)

theorem mem_wayBelowUp_ofFilter_of_round {f : Filter (D := D)} (hr : IsRound f) {a : D} :
  f.mem (wayBelowUp a) <-> a << ofFilter f := by
  constructor
  * intro ha
    obtain <b, hab, hb> := hr ha
    exact (wayBelow_sSup_iff (codes_nonempty f) (codes_directed f)).2 <b, hb, hab>
  * intro ha
    obtain <b, hbS, hab> := (wayBelow_sSup_iff (codes_nonempty f) (codes_directed f)).1 ha
    exact f.up_mem hbS <a, rfl> (wayBelowUp_anti hab.le)

theorem toFilter_ofFilter {f : Filter (D := D)} (hr : IsRound f) :
  toFilter (ofFilter f) = f := by
  refine NeighborhoodSystem.Element.ext (V := toNeighborhoodSystem) fun U => ?_
  constructor
  * intro hU
    obtain <a, <rfl, ha>> := hU
    exact (mem_wayBelowUp_ofFilter_of_round hr).2 ha

```

```

* intro hU
  obtain <a, rfl> := f.sub hU
  refine <a, <rfl, ?_>>
  exact (mem_wayBelowUp_ofFilter_of_round hr).1 hU

section Continuous

variable (hD : IsContinuousLattice D)
include hD

theorem toFilter_isRound (x : D) : IsRound (toFilter x) := by
  intro a ha
  have ha' : a << x := mem_wayBelowUp_toFilter.mp ha
  obtain <b, hab, hbx> := wayBelow_interpolate hD ha'
  exact <b, hab, mem_wayBelowUp_toFilter.mpr hbx>

theorem ofFilter_toFilter (x : D) : ofFilter (toFilter x) = x := by
  have hset : codes (toFilter x) = {a : D | a << x} := by
    ext a
    exact mem_wayBelowUp_toFilter
  change sSup (codes (toFilter x)) = x
  rw [hset, hD.sSup_wayBelow x]

/-- Round filters of the `Up`-system. -/
abbrev RoundFilter : Type _ :=
  { f : Filter (D := D) // IsRound f }

/-- Order-isomorphism: continuous lattice <=> round `Up`-filters. -/
noncomputable def domainOrderIso : D ?o RoundFilter (D := D) where
  toFun x := <toFilter x, toFilter_isRound hD x>
  invFun f := ofFilter (D := D) f.1
  left_inv x := ofFilter_toFilter hD x
  right_inv f := Subtype.ext (toFilter_ofFilter f.2)
  map_rel_iff' := by
    intro x y
    constructor
    * intro h
      have : ofFilter (toFilter x) <= ofFilter (toFilter y) := by
        refine sSup_le fun a ha => ?_
        have ha' : a << x := mem_wayBelowUp_toFilter.mp ha
        exact le_sSup (h (wayBelowUp a) <a, <rfl, ha'>>)
      simpa [ofFilter_toFilter hD] using this
    * intro hxy
      exact toFilter_mono hxy

/-- Order-embedding into all filters (forgets roundness). -/
noncomputable def domainEmbedding :
  D ?o Filter (D := D) where
  toFun := toFilter
  inj' := by
    intro x y h
    rw [ <- ofFilter_toFilter hD x, <- ofFilter_toFilter hD y, h ]
  map_rel_iff' := by
    intro x y
    constructor
    * intro h
      have : ofFilter (toFilter x) <= ofFilter (toFilter y) := by
        refine sSup_le fun a ha => ?_
        have ha' : a << x := mem_wayBelowUp_toFilter.mp ha
        exact le_sSup (h (wayBelowUp a) <a, <rfl, ha'>>)
      simpa [ofFilter_toFilter hD] using this
    * exact fun hxy => toFilter_mono hxy

```

```

end Continuous

end ContinuousLatticeToNeighborhood

/-- Blueprint name. -/
abbrev continuousLattice_to_neighborhoodSystem {D : Type*} [CompleteLattice D]
  (_hD : IsContinuousLattice D) : NeighborhoodSystem D :=
  ContinuousLatticeToNeighborhood.toNeighborhoodSystem

end ScottModels

```

A.5 ScottModels/InfoSysToIdealCompletion.lean

183 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Mathlib.Order.Ideal
import Scott1982.Factoid45

/-!
# Information systems -> ideal completion (1982 -> algebraic dcpo)

The domain  $|A|$  of an information system is the ideal completion of its
poset of finite elements  $u$  (closures of consistent finite token sets):
 $A.Element \rightarrow_o Order.Ideal (FiniteElement A)$ .

Packaging of Factoids 4.4-4.5 ( $\text{directedSup}$ ,  $\text{algebraicity}$ ,  $\text{compact\_closure}$ ).
-/

namespace ScottModels

open Scott1982
open Scott1982.Constructive
open Order

namespace InfoSysToIdealCompletion

variable {a : Type*} [DecidableEq a] (A : InfoSys a)

/-- Finite (compact) elements: closures of consistent finite token sets. -/
abbrev FiniteElement : Type _ :=
  { x : A.Element // exists (u : Finset a) (hu : u in A.Con), x = A.closure u hu }

theorem isFinite_bot :
  exists (u : Finset a) (hu : u in A.Con), A.botElement = A.closure u hu :=
  <{ }, A.con_empty, A.botElement_eq_closure_empty>

/-- Bottom, as a finite element. -/
def botFinite : FiniteElement A := <A.botElement, isFinite_bot A>

/-- Finite approximants of  $x$ , as  $FiniteElement$ 's. -/
def finiteApproximants (x : A.Element) : Set (FiniteElement A) :=
  { y | (y : A.Element) in A.finiteApproximants x }

```

```

theorem mem_finiteApproximants_of_le {x : A.Element} {y : FiniteElement A}
  (hle : (y : A.Element) <= x) : y in finiteApproximants A x := by
  rcases y with <yval, <u, hu, rfl>>
  refine <u, hu, ?_, rfl>
  intro a ha
  exact hle (A.subset_closure hu ha)

theorem le_of_mem_finiteApproximants {x : A.Element} {y : FiniteElement A}
  (hy : y in finiteApproximants A x) : (y : A.Element) <= x := by
  obtain <u, hu, huX, hyeq> := hy
  exact hyeq |> A.closure_le_element x hu huX

theorem mem_finiteApproximants_iff {x : A.Element} {y : FiniteElement A} :
  y in finiteApproximants A x <-> (y : A.Element) <= x :=
  <le_of_mem_finiteApproximants A, mem_finiteApproximants_of_le A>

theorem nonempty_finiteApproximants (x : A.Element) :
  (finiteApproximants A x).Nonempty :=
  <botFinite A, mem_finiteApproximants_of_le A (A.botElement_le x)>

theorem directed_finiteApproximants (x : A.Element) :
  DirectedOn (* <= *) (finiteApproximants A x) := by
  intro y_1 hy_1 y_2 hy_2
  obtain <u, hu, huX, hy_1eq> := hy_1
  obtain <v, hv, hvX, hy_2eq> := hy_2
  obtain <w, hw, hwX, huw, hvw> := A.closures_directed x hu hv huX hvX
  refine <<A.closure w hw, <w, hw, rfl>>, <w, hw, hwX, rfl>, ?_, ?_>
  * -- y_1 <= closure w
    change (y_1 : A.Element) <= A.closure w hw
    exact hy_1eq |> huw
  * change (y_2 : A.Element) <= A.closure w hw
    exact hy_2eq |> hvw

theorem isLowerSet_finiteApproximants (x : A.Element) :
  IsLowerSet (finiteApproximants A x) := by
  intro y_1 y_2 hle hy_2
  exact mem_finiteApproximants_of_le A (le_trans hle (le_of_mem_finiteApproximants A hy_2))

/-- Ideal of finite approximants of `x`. -/
def toIdeal (x : A.Element) : Ideal (FiniteElement A) where
  carrier := finiteApproximants A x
  lower' := isLowerSet_finiteApproximants A x
  nonempty' := nonempty_finiteApproximants A x
  directed' := directed_finiteApproximants A x

theorem mem_toIdeal_iff {x : A.Element} {y : FiniteElement A} :
  y in toIdeal A x <-> (y : A.Element) <= x :=
  mem_finiteApproximants_iff A

/-- Underlying set of elements of an ideal of finite elements. -/
def idealCarrier (I : Ideal (FiniteElement A)) : Set A.Element :=
  Subtype.val '' (I : Set (FiniteElement A))

theorem nonempty_idealCarrier (I : Ideal (FiniteElement A)) :
  (idealCarrier A I).Nonempty := by
  obtain <y, hy> := I.nonempty
  exact <y, y, hy, rfl>

theorem directed_idealCarrier (I : Ideal (FiniteElement A)) :
  A.IsDirected (idealCarrier A I) := by
  intro x y hx hy
  obtain <x', hx', rfl> := hx

```

```

obtain <y', hy', rfl> := hy
obtain <z', hz', hxz, hyz> := I.directed x' hx' y' hy'
exact <z', <z', hz', rfl>, hxz, hyz>

/-- Retraction: directed supremum of the finite elements in the ideal. -/
noncomputable def ofIdeal (I : Ideal (FiniteElement A)) : A.Element :=
  A.directedSup (idealCarrier A I) (nonempty_idealCarrier A I) (directed_idealCarrier A I)

theorem le_ofIdeal_of_mem {I : Ideal (FiniteElement A)} {y : FiniteElement A}
  (hy : y in I) : (y : A.Element) <= ofIdeal A I :=
  A.le_directedSup _ _ _ <y, hy, rfl>

theorem ofIdeal_toIdeal (x : A.Element) : ofIdeal A (toIdeal A x) = x := by
  -- idealCarrier (toIdeal x) = finiteApproximants as Elements = A.finiteApproximants x
  have hset : idealCarrier A (toIdeal A x) = A.finiteApproximants x := by
    ext z
    constructor
    * intro hz
      obtain <y, hy, rfl> := hz
      exact hy
    * intro hz
      refine <<z, ?_>, hz, rfl>
      obtain <u, hu, _, rfl> := hz
      exact <u, hu, rfl>
  -- directedSup of that set is x
  change A.directedSup (idealCarrier A (toIdeal A x)) _ _ = x
  simp_rw [hset]
  exact (A.eq_directedSup_finiteApproximants x).symm

theorem toIdeal_ofIdeal (I : Ideal (FiniteElement A)) : toIdeal A (ofIdeal A I) = I := by
  refine Ideal.ext ?_
  ext y
  constructor
  * intro hy
    have hle : (y : A.Element) <= ofIdeal A I := le_of_mem_finiteApproximants A hy
    rcases y with <yval, <u, hu, rfl>>
    obtain <z, hz, hcle> :=
      A.compact_closure (idealCarrier A I) (nonempty_idealCarrier A I)
      (directed_idealCarrier A I) hu hle
    obtain <z', hz', rfl> := hz
    exact I.lower hcle hz'
  * intro hy
    exact mem_finiteApproximants_of_le A (le_ofIdeal_of_mem A hy)

theorem toIdeal_mono {x y : A.Element} (hxy : x <= y) : toIdeal A x <= toIdeal A y := by
  intro z hz
  exact mem_finiteApproximants_of_le A
    (le_trans (le_of_mem_finiteApproximants A hz) hxy)

/-- Order isomorphism: domain elements <-> ideals of finite elements. -/
noncomputable def domainOrderIso : A.Element ?o Ideal (FiniteElement A) where
  toFun := toIdeal A
  invFun := ofIdeal A
  left_inv := ofIdeal_toIdeal A
  right_inv := toIdeal_ofIdeal A
  map_rel_iff' := by
    intro x y
    constructor
    * intro h
      -- x <= y from toIdeal x <= toIdeal y via algebraicity
      rw [ <- ofIdeal_toIdeal A x, <- ofIdeal_toIdeal A y ]
      refine A.directedSup_le _ _ _ ?_
      intro z hz

```

```

    obtain <z', hz', rfl> := hz
    exact le_ofIdeal_of_mem A (h hz')
    * exact toIdeal_mono A

end InfoSysToIdealCompletion

/-- Blueprint name: `|A|` as the ideal completion of its finite elements. -/
noncomputable abbrev infoSys_to_idealCompletion {a : Type*} [DecidableEq a] (A : InfoSys a) :
  A.Element ?o Ideal (InfoSysToIdealCompletion.FiniteElement A) :=
  InfoSysToIdealCompletion.domainOrderIso A

end ScottModels

```

A.6 ScottModels/IdealCompletionToContinuousLattice.lean

78 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.CompleteLattice.Basic
import Mathlib.Order.Directed
import Scott1972.ContinuousLattice.WayBelow

/-!
# Algebraic complete lattices -> continuous lattices (ideal-completion -> 1972)

An algebraic complete lattice -- every element is the directed supremum of the
compact elements below it -- is a continuous lattice in Scott's sense
(`IsContinuousLattice`). Compact elements are way-below anything they lie under,
via Scott-openness of `Set.Ici k`.

This is the classical frontier of this article's blueprint
(`idealCompletion_to_continuousLattice`): Scott's `<<` is topological.
-/

namespace ScottModels

open Scott1972.ContinuousLattice
open scoped Scott1972.ContinuousLattice

namespace IdealCompletionToContinuousLattice

variable {D : Type*} [CompleteLattice D]

/-- Order-theoretic compactness: inaccessible by nonempty directed suprema. -/
def IsCompactElement (k : D) : Prop :=
  forall ?S : Set D?, S.Nonempty -> DirectedOn (* <= *) S -> k <= sSup S -> exists
    s in S, k <= s

/-- Compactness implies `Set.Ici k` is Scott-open. -/
theorem scottOpen_Ici_of_compact {k : D} (hk : IsCompactElement k) :
  ScottOpen (Set.Ici k) := by
  refine <isUpperSet_Ici k, fun S hS hSdir hmem => ?_>
  obtain <s, hsS, hks> := hk hS hSdir (Set.mem_Ici.1 hmem)
  exact <s, hsS, Set.mem_Ici.2 hks>

```

```

/-- Compact elements are way below anything above them. -/
theorem compact_wayBelow {k y : D} (hk : IsCompactElement k) (hky : k <= y) : k << y :=
  <Set.Ici k, scottOpen_Ici_of_compact hk, Set.mem_Ici.2 hky, subset_refl>

/-- Algebraicity: every element is the directed lub of compact elements below it. -/
def IsAlgebraicLattice (D : Type*) [CompleteLattice D] : Prop :=
  forall y : D,
    let S := {x : D | IsCompactElement x /\ x <= y}
    S.Nonempty /\ DirectedOn (* <= *) S /\ sSup S = y

/-- **Blueprint:** algebraic complete lattice => continuous lattice (Scott Def. 2.3). -/
theorem isContinuousLattice_of_algebraic (hA : IsAlgebraicLattice D) :
  IsContinuousLattice D := by
  intro y
  obtain <hne, hdir, hsup> := hA y
  refine <?_, fun z hz => ?_>
  * -- y is an upper bound of `{x | x << y}`
  intro x hx
  exact hx.le
  * -- any upper bound of the way-below is >= y
  -- every compact <= y is << y, hence <= z
  have hle : sSup {x : D | IsCompactElement x /\ x <= y} <= z := by
    refine sSup.le fun x hx => ?_
    exact hz (compact_wayBelow hx.1 hx.2)
  exact hsup |> hle

end IdealCompletionToContinuousLattice

/-- Blueprint name. -/
abbrev idealCompletion_to_continuousLattice {D : Type*} [CompleteLattice D]
  (hA : IdealCompletionToContinuousLattice.IsAlgebraicLattice D) :
  IsContinuousLattice D :=
  IdealCompletionToContinuousLattice.isContinuousLattice_of_algebraic hA

end ScottModels

```

A.7 ScottModels/PresentationDomains.lean

178 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import ScottModels.InfoSysToNeighborhood
import ScottModels.InfoSysToIdealCompletion
import ScottModels.NeighborhoodToInfoSys
import ScottModels.ContinuousLatticeToNeighborhood
import ScottModels.IdealCompletionToContinuousLattice

/-!
# Presentation domain equivalences

## InfoSys / neighbourhood / ideal (constructive)

```

```

For any information system `A`:
`|A|` ?o `basic-open` neighbourhood filters ?o `Ideal (FiniteElement A)`.

Under a decidable `NbhdBasis`, the same triangle starts from `|?|`.

## Continuous lattices (1972 corner)

Points of a continuous lattice `D` are round `Up`-filters, not arbitrary filters:
`D` ?o `RoundFilter`. With `DecidableEq D`, the `Up`-system admits an `NbhdBasis`,
so round filters transport to a subtype of the induced InfoSys domain (and thence
into the ideal-completion triangle). The full (non-round) `|?|` / `|A|` is properly
larger.
-/

namespace ScottModels

open Scott1982
open Scott1972.ContinuousLattice
open Scott1980.Neighborhood
open Order
open scoped Scott1972.ContinuousLattice
open ContinuousLatticeToNeighborhood

section InfoSysTriangle
variable {a : Type*} [DecidableEq a]

/-- Neighbourhood filters of `|A|` <-> ideals of finite elements (via `|A|`). -/
noncomputable def neighborhood_ideal_iso (A : InfoSys a) :
  (InfoSysToNeighborhood.toNeighborhoodSystem A).Element ?o
  Ideal (InfoSysToIdealCompletion.FiniteElement A) :=
  (InfoSysToNeighborhood.domainOrderIso A).symm.trans
  (InfoSysToIdealCompletion.domainOrderIso A)

variable {? b : Type*} [DecidableEq ?]

/-- Under a decidable neighbourhood basis, `|?|` <-> InfoSys domain <-> ideal completion.
-/
noncomputable def nbhdBasis_ideal_iso (B : NbhdBasis ? b) :
  B.system.Element ?o Ideal (InfoSysToIdealCompletion.FiniteElement B.toInfoSys) :=
  B.domainOrderIso.trans (InfoSysToIdealCompletion.domainOrderIso B.toInfoSys)

/-- Constructive 1980 <-> 1982 <-> ideal triangle for InfoSys domains. -/
noncomputable abbrev presentation_domains_equiv_infoSys (A : InfoSys a) :=
  neighborhood_ideal_iso A

end InfoSysTriangle

/-! ## Continuous lattice presentations via round `Up`-filters -/

universe u
variable {D : Type u} [CompleteLattice D] [DecidableEq D]

/-- Decidable coding of the `Up`-neighbourhood system: tokens are elements of `D`. -/
def wayBelowNbhdBasis : NbhdBasis D D where
  system := toNeighborhoodSystem
  nbhd := wayBelowUp
  nbhd_mem := fun a => <a, rfl>
  exhaustive := by
    intro X hX
    obtain <a, rfl> := hX
    exact <a, rfl>
  botIdx := False
  botIdx_eq := wayBelowUp_bot

```

```

theorem wayBelowNbhdBasis_system :
  (wayBelowNbhdBasis (D := D)).system = toNeighborhoodSystem :=
  rfl

/-- Roundness transported to InfoSys elements of the `Up`-basis. -/
def IsRoundInfoSysElement (e : (wayBelowNbhdBasis (D := D)).toInfoSys.Element) : Prop :=
  IsRound ((wayBelowNbhdBasis (D := D)).domainOrderIso.symm e)

abbrev RoundInfoSysElement : Type _ :=
  { e : (wayBelowNbhdBasis (D := D)).toInfoSys.Element // IsRoundInfoSysElement (D := D) e }

/-- Round `Up`-filters <-> round elements of the coded InfoSys. -/
noncomputable def roundFilter_infoSys_iso :
  RoundFilter (D := D) ?o RoundInfoSysElement (D := D) where
  toFun := fun f =>
    <(wayBelowNbhdBasis (D := D)).domainOrderIso f.1, by
      let f' : (wayBelowNbhdBasis (D := D)).system.Element := f.1
      change IsRound ((wayBelowNbhdBasis (D := D)).domainOrderIso.symm
        ((wayBelowNbhdBasis (D := D)).domainOrderIso f'))
      rw [OrderIso.symm_apply_apply]
      exact f.2>
  invFun := fun e =>
    <(wayBelowNbhdBasis (D := D)).domainOrderIso.symm e.1, e.2>
  left_inv := fun f => Subtype.ext <|
    (wayBelowNbhdBasis (D := D)).domainOrderIso.left_inv f.1
  right_inv := fun e => Subtype.ext <|
    (wayBelowNbhdBasis (D := D)).domainOrderIso.right_inv e.1
  map_rel_iff' := by
    intro f g
    exact (wayBelowNbhdBasis (D := D)).domainOrderIso.map_rel_iff

section Continuous

variable (hD : IsContinuousLattice D)
include hD

/-- **1972 <-> round 1980:** continuous lattice <-> round `Up`-filters. -/
noncomputable def continuousLattice_roundFilter_iso :
  D ?o RoundFilter (D := D) :=
  ContinuousLatticeToNeighborhood.domainOrderIso hD

/-- **1972 <-> round 1982:** continuous lattice <-> round InfoSys elements of the `Up`-
basis. -/
noncomputable def continuousLattice_roundInfoSys_iso :
  D ?o RoundInfoSysElement (D := D) :=
  (continuousLattice_roundFilter_iso hD).trans roundFilter_infoSys_iso

/-- Round InfoSys elements <-> ideals of finite elements that come from round InfoSys
elements (via the ideal-completion iso). -/
noncomputable def roundInfoSys_ideal_iso :
  RoundInfoSysElement (D := D) ?o
  { I : Ideal (InfoSysToIdealCompletion.FiniteElement
    (wayBelowNbhdBasis (D := D)).toInfoSys) //
    IsRoundInfoSysElement (D := D)
    ((InfoSysToIdealCompletion.domainOrderIso
      (wayBelowNbhdBasis (D := D)).toInfoSys).symm I) } := by
  let A := (wayBelowNbhdBasis (D := D)).toInfoSys
  let ?E := InfoSysToIdealCompletion.domainOrderIso A
  refine {
    toFun := fun e => <?E e.1, by
      change IsRoundInfoSysElement (?E.symm (?E e.1))
      rw [OrderIso.symm_apply_apply]

```

```

    exact e.2>
  invFun := fun I => <?E.symm I.1, I.2>
  left_inv := fun e => Subtype.ext (?E.left_inv e.1)
  right_inv := fun I => Subtype.ext (?E.right_inv I.1)
  map_rel_iff' := by
    intro e_1 e_2
    exact ?E.map_rel_iff
}

/-- **Blueprint:** three-presentation equivalence for continuous lattices.

`D ?o RoundFilter ?o RoundInfoSysElement`, with the `Up`-system?s `NbhdBasis`
supplying the 1980 <-> 1982 coding. (Raw `|?|` / full `|A|` remain larger.) -/
noncomputable def presentation_domains_equiv :
  D ?o RoundInfoSysElement (D := D) :=
  continuousLattice_roundInfoSys_iso hD

/-- The three presentations determine the same domain (round corner). -/
theorem exists_presentation_domains_equiv :
  Nonempty (D ?o RoundInfoSysElement (D := D)) :=
  <presentation_domains_equiv hD>

/-- Extended form through ideal completion of finite elements of the `Up`-InfoSys. -/
noncomputable def presentation_domains_equiv_ideal :
  D ?o
  { I : Ideal (InfoSysToIdealCompletion.FiniteElement
    (wayBelowNbhdBasis (D := D)).toInfoSys) //
    IsRoundInfoSysElement (D := D)
    ((InfoSysToIdealCompletion.domainOrderIso
      (wayBelowNbhdBasis (D := D)).toInfoSys).symm I) } :=
  (presentation_domains_equiv hD).trans (roundInfoSys_ideal_iso (D := D))

end Continuous

end ScottModels

```

A.8 ScottModels/InfoSysConstructions.lean

495 lines.

Lean 4 source (lines 1–400)

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Mathlib.Data.Sum.Order
import Mathlib.Order.WithBot
import Scott1982.Theorem72
import Scott1982.Proposition64

/-!
# Construction equivalence -- products, separated sums, and function spaces

* **Product:** `|A x B| ?o |A| x |B|` via `pairElements` / projections (Prop 6.2).
* **Separated sum:** `|A + B| ?o WithBot (|A| ? |B|)` via `inl`/`inr` (Prop 6.4).
  Classification of sum elements uses classical case-split on token polarity
  (`Classical.choice` in the footprint).
* **Function space:** `|A -> B| ?o ApproximableMap A B` via Theorem 7.2

```

```

`approxMap_toElement` / `element_toApproxMap`.

1972 counterpart: products of continuous lattices (`proposition_2_9_a`); function space
Thm 3.3 in the sibling package.
-/

namespace ScottModels

open Scott1982
open Scott1982.Constructive
open Scott1982.InfoSys
open Scott1982.InfoSys.ApproximableMap

namespace InfoSysConstructions

variable {a b : Type*} [DecidableEq a] [DecidableEq b] (A : InfoSys a) (B : InfoSys b)

/-- Split a product-domain element into its two projections. -/
def unpair (z : (productSystem A B).Element) : A.Element x B.Element :=
  ((fstMap A B).toElement z, (sndMap A B).toElement z)

theorem unpair_pairElements (x : A.Element) (y : B.Element) :
  unpair A B (pairElements A B x y) = (x, y) := by
  simp [unpair, fstMap_pairElements, sndMap_pairElements]

theorem pairElements_unpair (z : (productSystem A B).Element) :
  pairElements A B ((fstMap A B).toElement z) ((sndMap A B).toElement z) = z :=
  element_eq_of_fst_snd A B _ z (fstMap_pairElements A B _ _) (sndMap_pairElements A B _ _)

theorem pairElements_mono {x_1 x_2 : A.Element} {y_1 y_2 : B.Element}
  (hx : x_1 <= x_2) (hy : y_1 <= y_2) :
  pairElements A B x_1 y_1 <= pairElements A B x_2 y_2 := by
  intro p hp
  exact <fun hbot => hx (hp.1 hbot), fun hbot => hy (hp.2 hbot)>

theorem unpair_mono {z_1 z_2 : (productSystem A B).Element} (hz : z_1 <= z_2) :
  unpair A B z_1 <= unpair A B z_2 := by
  constructor
  * exact (fstMap A B).toElement_mono hz
  * exact (sndMap A B).toElement_mono hz

/-- **Product domain iso (1982).** `|A x B| ?o |A| x |B|`. -/
noncomputable def productDomainIso :
  A.Element x B.Element ?o (productSystem A B).Element where
  toFun := fun p => pairElements A B p.1 p.2
  invFun := unpair A B
  left_inv := fun p => unpair_pairElements A B p.1 p.2
  right_inv := pairElements_unpair A B
  map_rel_iff' := by
    intro p q
    constructor
    * intro h
      change pairElements A B p.1 p.2 <= pairElements A B q.1 q.2 at h
      constructor
      * have hp := (fstMap A B).toElement_mono h
        rw [fstMap_pairElements A B p.1 p.2, fstMap_pairElements A B q.1 q.2] at hp
        exact hp
      * have hp := (sndMap A B).toElement_mono h
        rw [sndMap_pairElements A B p.1 p.2, sndMap_pairElements A B q.1 q.2] at hp
        exact hp
    * intro h
      exact pairElements_mono A B h.1 h.2

```

```

/-! ## Separated sum `|A + B| ?o WithBot (|A| ? |B|)` -/

private theorem rhtFinset_singleton_left (x : a) :
  rhtFinset ({SumToken.left x} : Finset (SumToken a b)) = {} := by
  ext y
  constructor
  * intro hy
    have : SumToken.right y in ({SumToken.left x} : Finset _) := (mem_rhtFinset).1 hy
    exact False.elim (nomatch Finset.mem_singleton.mp this)
  * intro hy
    exact False.elim (Finset.notMem_empty y hy)

private theorem lftFinset_singleton_right (y : b) :
  lftFinset ({SumToken.right y} : Finset (SumToken a b)) = {} := by
  ext x
  constructor
  * intro hx
    have : SumToken.left x in ({SumToken.right y} : Finset _) := (mem_lftFinset).1 hx
    exact False.elim (nomatch Finset.mem_singleton.mp this)
  * intro hx
    exact False.elim (Finset.notMem_empty x hx)

theorem left_bot_mem_inlMap_toElement (x : A.Element) :
  SumToken.left A.bot in ((inlMap A B).toElement x).carrier := by
  refine <{A.bot}, ?_, ?_>
  * intro a ha
    have : a = A.bot := Finset.mem_singleton.mp (Finset.mem_coe.1 ha)
    subst this
    exact factoid_3_2 A x
  * refine <A.con_sing A.bot, ?_, rhtFinset_singleton_left (b := b) A.bot, ?_>
  * exact Or.inl <by rw [lftFinset_singleton_left]; exact A.con_sing A.bot,
    rhtFinset_singleton_left (b := b) A.bot>
  * rw [lftFinset_singleton_left]
    exact proposition_2_3_iii A (A.con_sing A.bot)

theorem right_bot_mem_inrMap_toElement (y : B.Element) :
  SumToken.right B.bot in ((inrMap A B).toElement y).carrier := by
  refine <{B.bot}, ?_, ?_>
  * intro b hb
    have : b = B.bot := Finset.mem_singleton.mp (Finset.mem_coe.1 hb)
    subst this
    exact factoid_3_2 B y
  * refine <B.con_sing B.bot, ?_, lftFinset_singleton_right (a := a) B.bot, ?_>
  * exact Or.inr <lftFinset_singleton_right (a := a) B.bot,
    by rw [rhtFinset_singleton_right]; exact B.con_sing B.bot>
  * rw [rhtFinset_singleton_right]
    exact proposition_2_3_iii B (B.con_sing B.bot)

theorem sumElementLft_inlMap_toElement (x : A.Element) :
  sumElementLft A B ((inlMap A B).toElement x) A.bot
  (left_bot_mem_inlMap_toElement A B x) = x := by
  apply le_antisymm
  * intro a ha
    -- ha : left a in inl(x)
    obtain <u, hu, <huCon, hSum, hr, hEnt>> := ha
    have hEntA : A.Ent u a := by
      have : lftFinset ({SumToken.left a} : Finset (SumToken a b)) = {a} :=
        lftFinset_singleton_left a
      simpa [this] using hEnt a (Finset.mem_singleton_self a)
    exact x.closed u a hu hEntA
  * intro a ha
    change SumToken.left a in ((inlMap A B).toElement x).carrier
    refine <{a}, ?_, ?_>

```

```

* intro b hb
  have : b = a := Finset.mem_singleton.mp (Finset.mem_coe.1 hb)
  subst this; exact ha
* refine <A.con_sing a, Or.inl <by rw [lftFinset_singleton_left]; exact A.con_sing a,
  rhtFinset_singleton_left (b := b) a>, rhtFinset_singleton_left (b := b) a, ?_>
  rw [lftFinset_singleton_left]
  exact proposition_2_3_iii A (A.con_sing a)

theorem sumElementRht_inrMap_toElement (y : B.Element) :
  sumElementRht A B ((inrMap A B).toElement y) B.bot
  (right_bot_mem_inrMap_toElement A B y) = y := by
  apply le_antisymm
* intro b hb
  obtain <w, hw, <hwCon, hSum, hl, hEnt>> := hb
  have hEntB : B.Ent w b := by
    have : rhtFinset ({SumToken.right b} : Finset (SumToken a b)) = {b} :=
      rhtFinset_singleton_right b
    simp [this] using hEnt b (Finset.mem_singleton_self b)
  exact y.closed w b hw hEntB
* intro b hb
  change SumToken.right b in ((inrMap A B).toElement y).carrier
  refine <{b}, ?_, ?_>
* intro c hc
  have : c = b := Finset.mem_singleton.mp (Finset.mem_coe.1 hc)
  subst this; exact hb
* refine <B.con_sing b, Or.inr <lftFinset_singleton_right (a := a) b,
  by rw [rhtFinset_singleton_right]; exact B.con_sing b>,
  lftFinset_singleton_right (a := a) b, ?_>
  rw [rhtFinset_singleton_right]
  exact proposition_2_3_iii B (B.con_sing b)

theorem inlMap_toElement_injective :
  Function.Injective (inlMap A B).toElement := by
  intro x y h
  refine le_antisymm ?_ ?_
* intro a ha
  have hx := sumElementLft_inlMap_toElement A B x
  have : SumToken.left a in ((inlMap A B).toElement x).carrier := by
    rw [← hx] at ha; exact ha
  rw [h] at this
  have hy := sumElementLft_inlMap_toElement A B y
  have : a in (sumElementLft A B ((inlMap A B).toElement y) A.bot
    (left_bot_mem_inlMap_toElement A B y)).carrier := this
  rwa [hy] at this
* intro a ha
  have hy := sumElementLft_inlMap_toElement A B y
  have : SumToken.left a in ((inlMap A B).toElement y).carrier := by
    rw [← hy] at ha; exact ha
  rw [← h] at this
  have hx := sumElementLft_inlMap_toElement A B x
  have : a in (sumElementLft A B ((inlMap A B).toElement x) A.bot
    (left_bot_mem_inlMap_toElement A B x)).carrier := this
  rwa [hx] at this

theorem inrMap_toElement_injective :
  Function.Injective (inrMap A B).toElement := by
  intro x y h
  refine le_antisymm ?_ ?_
* intro b hb
  have hx := sumElementRht_inrMap_toElement A B x
  have : SumToken.right b in ((inrMap A B).toElement x).carrier := by
    rw [← hx] at hb; exact hb
  rw [h] at this

```

```

have hy := sumElementRht_inrMap_toElement A B y
have : b in (sumElementRht A B ((inrMap A B).toElement y) B.bot
  (right_bot_mem_inrMap_toElement A B y)).carrier := this
rwa [hy] at this
* intro b hb
have hy := sumElementRht_inrMap_toElement A B y
have : SumToken.right b in ((inrMap A B).toElement y).carrier := by
  rw [← hy] at hb; exact hb
rw [← h] at this
have hx := sumElementRht_inrMap_toElement A B x
have : b in (sumElementRht A B ((inrMap A B).toElement x) B.bot
  (right_bot_mem_inrMap_toElement A B x)).carrier := this
rwa [hx] at this

/-- Every sum element is False, a pure left copy, or a pure right copy. Classical. -/
theorem sum_element_trichotomy (z : (sumSystem A B).Element) :
  z = (sumSystem A B).botElement \ /
    (exists x : A.Element, z = (inlMap A B).toElement x) \ /
    (exists y : B.Element, z = (inrMap A B).toElement y) := by
  classical
  by_cases hL : exists x : a, SumToken.left x in z.carrier
  * obtain <x0, hx0> := hL
    exact Or.inr (Or.inl <sumElementLft A B z x0 hx0, (inlMap_toElement_sumElementLft A B z x0
      hx0).symm>)
  * by_cases hR : exists y : b, SumToken.right y in z.carrier
    * obtain <y0, hy0> := hR
      exact Or.inr (Or.inr <sumElementRht A B z y0 hy0, (inrMap_toElement_sumElementRht A B z
        y0 hy0).symm>)
    * exact Or.inl (eq_botElement_of_no_injections A B z
      (fun x hx => hL <x, hx>) (fun y hy => hR <y, hy>))

/-- Assemble a sum-domain element from a separated-sum code. -/
def assembleSum : WithBot (A.Element ? B.Element) -> (sumSystem A B).Element
| False => (sumSystem A B).botElement
| some (.inl x) => (inlMap A B).toElement x
| some (.inr y) => (inrMap A B).toElement y

/-- Every sum-domain element has a separated-sum code. -/
private theorem exists_assembleSum_eq (z : (sumSystem A B).Element) :
  exists w : WithBot (A.Element ? B.Element), assembleSum A B w = z := by
  rcases sum_element_trichotomy A B z with hbot | <x, hx> | <y, hy>
  * exact <False, hbot.symm>
  * exact <some (.inl x), hx.symm>
  * exact <some (.inr y), hy.symm>

/-- Classify a sum-domain element as `WithBot (|A| ? |B|)`. Classical. -/
noncomputable def classifySum (z : (sumSystem A B).Element) :
  WithBot (A.Element ? B.Element) :=
  Classical.choose (exists_assembleSum_eq A B z)

theorem assembleSum_classifySum (z : (sumSystem A B).Element) :
  assembleSum A B (classifySum A B z) = z :=
  Classical.choose_spec (exists_assembleSum_eq A B z)

theorem not_left_mem_sum_botElement {x : a} :
  SumToken.left x notin ((sumSystem A B).botElement).carrier := by
  intro hx
  have hEnt : (sumSystem A B).Ent {sumBot} (.left x) := hx
  -- Ent {bot} (left x) requires lftFinset {bot} != {}
  rcases hEnt with <_, <hne, _>>
  exact hne lftFinset_singleton_bot

theorem not_right_mem_sum_botElement {y : b} :

```

```

    SumToken.right y notin ((sumSystem A B).botElement).carrier := by
intro hy
have hEnt : (sumSystem A B).Ent {sumBot} (.right y) := hy
rcases hEnt with <_, <hne, _>>
exact hne rhtFinset_singleton_bot

theorem sumElementLft_eq {z : (sumSystem A B).Element} {x0 x1 : a}
  (hx0 : SumToken.left x0 in z.carrier) (hx1 : SumToken.left x1 in z.carrier) :
  sumElementLft A B z x0 hx0 = sumElementLft A B z x1 hx1 := by
  refine le_antisymm ?_ ?_ <;> intro a ha <;> exact ha

theorem sumElementRht_eq {z : (sumSystem A B).Element} {y0 y1 : b}
  (hy0 : SumToken.right y0 in z.carrier) (hy1 : SumToken.right y1 in z.carrier) :
  sumElementRht A B z y0 hy0 = sumElementRht A B z y1 hy1 := by
  refine le_antisymm ?_ ?_ <;> intro b hb <;> exact hb

theorem assembleSum_injective :
  Function.Injective (assembleSum A B) := by
intro w_1 w_2 h
cases w_1 with
| bot =>
  cases w_2 with
  | bot => rfl
  | coe s_2 =>
    cases s_2 with
    | inl x =>
      change (sumSystem A B).botElement = (inlMap A B).toElement x at h
      have hx := left_bot_mem_inlMap_toElement A B x
      rw [← h] at hx
      exact False.elim (not_left_mem_sum_botElement A B hx)
    | inr y =>
      change (sumSystem A B).botElement = (inrMap A B).toElement y at h
      have hy := right_bot_mem_inrMap_toElement A B y
      rw [← h] at hy
      exact False.elim (not_right_mem_sum_botElement A B hy)
| coe s_1 =>
  cases w_2 with
  | bot =>
    cases s_1 with
    | inl x =>
      change (inlMap A B).toElement x = (sumSystem A B).botElement at h
      have hx := left_bot_mem_inlMap_toElement A B x
      rw [h] at hx
      exact False.elim (not_left_mem_sum_botElement A B hx)
    | inr y =>
      change (inrMap A B).toElement y = (sumSystem A B).botElement at h
      have hy := right_bot_mem_inrMap_toElement A B y
      rw [h] at hy
      exact False.elim (not_right_mem_sum_botElement A B hy)
  | coe s_2 =>
    cases s_1 with
    | inl x =>
      cases s_2 with
      | inl x' => exact congrArg (fun t => some (Sum.inl t)) (inlMap_toElement_injective A B
h)
      | inr y' =>
        change (inlMap A B).toElement x = (inrMap A B).toElement y' at h
        have hx := left_bot_mem_inlMap_toElement A B x
        rw [h] at hx
        exact False.elim <| not_mem_right_of_mem_left A B ((inrMap A B).toElement y')
          hx
          (right_bot_mem_inrMap_toElement A B y')
    | inr y =>

```

```

cases s_2 with
| inl x' =>
  change (inlMap A B).toElement y = (inlMap A B).toElement x' at h
  have hy := right_bot_mem_inlMap_toElement A B y
  rw [h] at hy
  exact False.elim <| not_mem_right_of_mem_left A B ((inlMap A B).toElement x')
    (left_bot_mem_inlMap_toElement A B x')
  hy
| inr y' => exact congrArg (fun t => some (Sum.inr t)) (inlMap_toElement_injective A B
h)

theorem classifySum_assembleSum (w : WithBot (A.Element ? B.Element)) :
  classifySum A B (assembleSum A B w) = w := by
  apply assembleSum_injective A B
  exact assembleSum_classifySum A B (assembleSum A B w)

theorem inlMap_toElement_le_iff {x y : A.Element} :
  (inlMap A B).toElement x <= (inlMap A B).toElement y <-> x <= y := by
  constructor
  * intro h a ha
  have : SumToken.left a in ((inlMap A B).toElement x).carrier := by
    -- from sumElementLft_inl round-trip carrier
    have hx := sumElementLft_inlMap_toElement A B x
    -- a in x = sumElementLft => left a in inl x
    have : a in (sumElementLft A B ((inlMap A B).toElement x) A.bot
      (left_bot_mem_inlMap_toElement A B x)).carrier := by
      simpa [hx] using ha
    exact this
  exact (sumElementLft_inlMap_toElement A B y) |>
    (show a in (sumElementLft A B ((inlMap A B).toElement y) A.bot
      (left_bot_mem_inlMap_toElement A B y)).carrier from h this)
  * exact (inlMap A B).toElement_mono

theorem inrMap_toElement_le_iff {x y : B.Element} :
  (inrMap A B).toElement x <= (inrMap A B).toElement y <-> x <= y := by
  constructor
  * intro h b hb
  have : SumToken.right b in ((inrMap A B).toElement x).carrier := by
    have hx := sumElementRht_inrMap_toElement A B x
    have : b in (sumElementRht A B ((inrMap A B).toElement x) B.bot
      (right_bot_mem_inrMap_toElement A B x)).carrier := by
      simpa [hx] using hb
    exact this
  exact (sumElementRht_inrMap_toElement A B y) |>
    (show b in (sumElementRht A B ((inrMap A B).toElement y) B.bot
      (right_bot_mem_inrMap_toElement A B y)).carrier from h this)
  * exact (inrMap A B).toElement_mono

theorem assembleSum_mono {w_1 w_2 : WithBot (A.Element ? B.Element)} (h : w_1 <= w_2) :
  assembleSum A B w_1 <= assembleSum A B w_2 := by
  cases w_1 with
  | bot => exact botElement_le _ _
  | coe s_1 =>
    cases w_2 with
    | bot => exact (WithBot.not_coe_le_bot _ h).elim
    | coe s_2 =>
      have hs : s_1 <= s_2 := WithBot.coe_le_coe.1 h
      cases s_1 with
      | inl x =>
        cases s_2 with
        | inl x' => exact (inlMap A B).toElement_mono (Sum.inl_le_inl_iff.1 hs)
        | inr y' => exact (Sum.not_inl_le_inr hs).elim
      | inr y =>

```

```

    cases s_2 with
    | inl x' => exact (Sum.not_inr_le_inl hs).elim
    | inr y' => exact (inrMap A B).toElement_mono (Sum.inr_le_inr_iff.1 hs)

/-- **Separated-sum domain iso (1982). ** `|A + B| ?o WithBot (|A| ? |B|)` . Classical. -/
Lean 4 source (lines 401–495)

noncomputable def sumDomainIso :
  WithBot (A.Element ? B.Element) ?o (sumSystem A B).Element where
toFun := assembleSum A B
invFun := classifySum A B
left_inv := classifySum_assembleSum A B
right_inv := assembleSum_classifySum A B
map_rel_iff' := by
  intro w_1 w_2
  constructor
  * intro h
  cases w_1 with
  | bot => exact bot_le
  | coe s_1 =>
    cases w_2 with
    | bot =>
      cases s_1 with
      | inl x =>
        exact False.elim (not_left_mem_sum_botElement A B
          (h (left_bot_mem_inlMap_toElement A B x)))
      | inr y =>
        exact False.elim (not_right_mem_sum_botElement A B
          (h (right_bot_mem_inrMap_toElement A B y)))
    | coe s_2 =>
      refine WithBot.coe_le_coe.2 ?_
      cases s_1 with
      | inl x =>
        cases s_2 with
        | inl x' => exact Sum.inl_le_inl_iff.2 ((inlMap_toElement_le_iff A B).1 h)
        | inr y' =>
          exact False.elim <|
            not_mem_right_of_mem_left A B ((inrMap A B).toElement y')
              (h (left_bot_mem_inlMap_toElement A B x))
              (right_bot_mem_inrMap_toElement A B y')
      | inr y =>
        cases s_2 with
        | inl x' =>
          exact False.elim <|
            not_mem_right_of_mem_left A B ((inlMap A B).toElement x')
              (left_bot_mem_inlMap_toElement A B x')
              (h (right_bot_mem_inrMap_toElement A B y))
        | inr y' => exact Sum.inr_le_inr_iff.2 ((inrMap_toElement_le_iff A B).1 h)
  * exact assembleSum_mono A B

/-! ## Function space `|A -> B| ?o ApproximableMap A B` -/

/-- Pointwise relation order on approximable maps (Prop 5.3 `Le` ). -/
instance instPartialOrderApproximableMap : PartialOrder (ApproximableMap A B) where
  le := @Le _ _ _ A B
  le_refl _ _ _ h := h
  le_trans _ _ _ hfg hgh _ _ hf := hgh (hfg hf)
  le_antisymm _ _ hfg hgf := ApproximableMap.ext fun _ _ => <fun h => hfg h, fun h => hgf h>

theorem approxMap_toElement_le_iff {f g : ApproximableMap A B} :
  approxMap_toElement A B f <= approxMap_toElement A B g <-> f <= g := by
  constructor
  * intro h u v hrel

```

```

    have hp : mkFunToken A B u v (f.rel_dom hrel) (f.rel_cod hrel) in
      (approxMap_toElement A B f).carrier :=
        (mem_approxMap_toElement A B f).2 hrel
    exact (mem_approxMap_toElement A B g).1 (h hp)
* intro hfg p hp
    exact (mem_approxMap_toElement A B g).2 (hfg ((mem_approxMap_toElement A B f).1 hp))

/-- **Function-space domain iso (1982, Thm 7.2).** `|A -> B| ?o ApproximableMap A B`. -/
noncomputable def functionSpaceDomainIso :
  ApproximableMap A B ?o (functionSystem A B).Element where
  toFun := approxMap_toElement A B
  invFun := element_toApproxMap A B
  left_inv := element_toApproxMap_approxMap_toElement A B
  right_inv := approxMap_toElement_element_toApproxMap A B
  map_rel_iff' := by
    intro f g
    exact approxMap_toElement_le_iff A B

end InfoSysConstructions

/-- Blueprint-facing name for the 1982 product domain isomorphism. -/
noncomputable abbrev infoSys_product_domain_equiv {a b : Type*} [DecidableEq a] [DecidableEq b]
  (A : InfoSys a) (B : InfoSys b) :
  A.Element x B.Element ?o (productSystem A B).Element :=
  InfoSysConstructions.productDomainIso A B

/-- Blueprint-facing name for the 1982 separated-sum domain isomorphism (classical). -/
noncomputable abbrev infoSys_sum_domain_equiv {a b : Type*} [DecidableEq a] [DecidableEq b]
  (A : InfoSys a) (B : InfoSys b) :
  WithBot (A.Element ? B.Element) ?o (sumSystem A B).Element :=
  InfoSysConstructions.sumDomainIso A B

/-- Blueprint-facing name for the 1982 function-space domain isomorphism. -/
noncomputable abbrev infoSys_function_space_domain_equiv {a b : Type*}
  [DecidableEq a] [DecidableEq b] (A : InfoSys a) (B : InfoSys b) :
  ApproximableMap A B ?o (functionSystem A B).Element :=
  InfoSysConstructions.functionSpaceDomainIso A B

end ScottModels

```

A.9 ScottModels/ScottMapBridge.lean

190 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Scott1972.ContinuousLattice.FunctionSpaces
import Scott1982.Factoid46
import Scott1982.Factoid36
import Scott1982.Proposition53
import ScottModels.InfoSysConstructions
import ScottModels.PresentationDomains

/-!
# Construction cross-links -- ApproximableMap <-> ScottContinuous <-> ScottMap

```

```

* **Factoid 4.6:** `ApproximableMap A B ?o ScottContinuous A B` (1982).
* **1972 transport:** `ScottMap D D'` conjugates along `presentation_domains_equiv`
  (or the round-filter iso) to pointwise-ordered maps on the round presentation.
* Blueprint packaging: `infoSys_constructions_equiv` bundles the three 1982 domain
  isos with these cross-links.
-/

namespace ScottModels

open Scott1982
open Scott1982.Constructive
open Scott1982.InfoSys
open Scott1982.InfoSys.ApproximableMap
open Scott1972.ContinuousLattice
open ContinuousLatticeToNeighborhood

/-! ## Factoid 4.6 as an order isomorphism -/

variable {a b : Type*} [DecidableEq a] [DecidableEq b]

theorem scottContinuous_ext {A : InfoSys a} {B : InfoSys b}
  {f g : ScottContinuous A B} (h : forall x, f.toFun x = g.toFun x) : f = g := by
  obtain <tf, mf, df> := f
  obtain <tg, mg, dg> := g
  have : tf = tg := funext h
  subst this
  rfl

/-- Pointwise order on Scott-continuous maps of InfoSys domains. -/
instance instPartialOrderScottContinuous (A : InfoSys a) (B : InfoSys b) :
  PartialOrder (ScottContinuous A B) where
  le f g := forall x, f.toFun x <= g.toFun x
  le_refl _ _ := le_refl _
  le_trans _ _ hfg hgh x := le_trans (hfg x) (hgh x)
  le_antisymm _ _ hfg hgf :=
    scottContinuous_ext fun x => le_antisymm (hfg x) (hgf x)

theorem ofScottContinuous_toScottContinuous {A : InfoSys a} {B : InfoSys b}
  (f : ApproximableMap A B) :
  ofScottContinuous (toScottContinuous f) = f := by
  refine ApproximableMap.ext fun u v => ?_
  constructor
  * intro <hu, hv, hsub>
    exact (f.rel_iff_closure_le hu hv).2
      (B.closure_le_element (f.toElement (A.closure u hu)) hv hsub)
  * intro hrel
    refine <f.rel_dom hrel, f.rel_cod hrel, ?_>
    intro y hy
    have hv : v in B.Con := f.rel_cod hrel
    have hu : u in A.Con := f.rel_dom hrel
    exact (f.rel_iff_closure_le hu hv).1 hrel (B.subset_closure hv (Finset.mem_coe.1 hy))

theorem toScottContinuous_ofScottContinuous {A : InfoSys a} {B : InfoSys b}
  (g : ScottContinuous A B) :
  toScottContinuous (ofScottContinuous g) = g :=
  scottContinuous_ext fun x => toElement_ofScottContinuous g x

/-- **Factoid 4.6.** Approximable maps <-> Scott-continuous maps of domains. -/
noncomputable def approximableMap_scottContinuous_equiv (A : InfoSys a) (B : InfoSys b) :
  ApproximableMap A B ?o ScottContinuous A B where
  toFun := toScottContinuous
  invFun := ofScottContinuous

```

```

left_inv := ofScottContinuous_toScottContinuous
right_inv := toScottContinuous_ofScottContinuous
map_rel_iff' := by
  intro f g
  exact (le_iff_toElement_le f g).symm

/-! ## ScottMap conjugation along the round presentation -/

variable {D E : Type*} [CompleteLattice D] [CompleteLattice E]

/-- Conjugate a Scott map along order isomorphisms of the underlying lattices. -/
noncomputable def conjScottMapFun {D' E' : Type*} [LE D'] [LE E']
  (?D : D →o D') (?E : E →o E') (f : ScottMap D E) : D' → E' :=
  ??E comp (f : D → E) comp ??D.symm

theorem conjScottMapFun_le_iff {D' E' : Type*} [PartialOrder D'] [PartialOrder E']
  (?D : D →o D') (?E : E →o E') {f g : ScottMap D E} :
  (forall x : D', conjScottMapFun ?D ?E f x ≤ conjScottMapFun ?D ?E g x) <-> f ≤ g
:= by
  constructor
  * intro h
  rw [ScottMap.le_def]
  intro x
  have hx : ?E (f x) ≤ ?E (g x) := by
    simpa [conjScottMapFun, OrderIso.symm_apply_apply] using h (?D x)
  exact ?E.map_rel_iff.mp hx
  * intro hfg x
  exact ?E.monotone (hfg (?D.symm x))

/-- Scott maps packaged as their conjugates along given presentation isos. -/
structure ConjScottMap {D' E' : Type*} [PartialOrder D'] [PartialOrder E']
  (?D : D →o D') (?E : E →o E') where
  scott : ScottMap D E

namespace ConjScottMap

variable {D' E' : Type*} [PartialOrder D'] [PartialOrder E']
variable (?D : D →o D') (?E : E →o E')

noncomputable instance : CoeFun (ConjScottMap ?D ?E) (fun _ => D' → E') where
  coe g := conjScottMapFun ?D ?E g.scott

theorem ext {f g : ConjScottMap ?D ?E} (h : f.scott = g.scott) : f = g := by
  cases f; cases g; congr

noncomputable instance : PartialOrder (ConjScottMap ?D ?E) where
  le f g := forall x, conjScottMapFun ?D ?E f.scott x ≤ conjScottMapFun ?D ?E g.scott x
  le_refl _ _ := le_refl _
  le_trans _ _ hfg hgh x := le_trans (hfg x) (hgh x)
  le_antisymm f g hfg hgf := by
    refine ext ?D ?E ?_
    exact le_antisymm
      ((conjScottMapFun_le_iff ?D ?E).1 hfg)
      ((conjScottMapFun_le_iff ?D ?E).1 hgf)

/-- **1972 <-> round presentation:** Scott maps ? conjugates along the given isos. -/
noncomputable def orderIso : ScottMap D E →o ConjScottMap ?D ?E where
  toFun f := <f>
  invFun g := g.scott
  left_inv _ := rfl
  right_inv _ := rfl
  map_rel_iff' := by
    intro f g

```

```

    exact conjScottMapFun_le_iff ?D ?E

end ConjScottMap

section ContinuousRoundFilter

variable (hD : IsContinuousLattice D) (hE : IsContinuousLattice E)
include hD hE

/-- Conjugation along round `Up`-filters (no `DecidableEq` needed). -/
noncomputable abbrev scottMap_roundFilter_iso :
  ScottMap D E ?o
  ConjScottMap (continuousLattice_roundFilter_iso hD)
    (continuousLattice_roundFilter_iso hE) :=
  ConjScottMap.orderIso (continuousLattice_roundFilter_iso hD)
    (continuousLattice_roundFilter_iso hE)

end ContinuousRoundFilter

section ContinuousRoundInfoSys

variable (hD : IsContinuousLattice D) (hE : IsContinuousLattice E)
variable [DecidableEq D] [DecidableEq E]
include hD hE

/-- Conjugation along round InfoSys elements of the `Up`-basis. -/
noncomputable abbrev scottMap_roundInfoSys_iso :
  ScottMap D E ?o
  ConjScottMap (presentation_domains_equiv hD) (presentation_domains_equiv hE) :=
  ConjScottMap.orderIso (presentation_domains_equiv hD) (presentation_domains_equiv hE)

end ContinuousRoundInfoSys

/-! ## Blueprint packaging

Bundled 1982 construction domain isos + 1972/1982 function-space cross-links. -/
namespace infoSys_constructions_equiv

noncomputable abbrev product := @infoSys_product_domain_equiv
noncomputable abbrev sum := @infoSys_sum_domain_equiv
noncomputable abbrev functionSpace := @infoSys_function_space_domain_equiv
noncomputable abbrev approximable_scottContinuous := @approximableMap_scottContinuous_equiv
noncomputable abbrev scottMap_roundFilter := @scottMap_roundFilter_iso
noncomputable abbrev scottMap_roundInfoSys := @scottMap_roundInfoSys_iso

end infoSys_constructions_equiv

end ScottModels

```

A.10 ScottModels/SexDomainEquation.lean

644 lines.

Lean 4 source (lines 1–400)

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic

```

```

import Scott1982.Factoid81

/-!
# Domain equation  $|T| \leq |A| + (|T| \times |T|)$  via unfolding

Factoid 8.1 presents  $T$  as an inductive token system whose official
right-hand side is  $\text{sumSystem } A \text{ (productSystem } T \text{ )}$ . The token map
 $\text{treeUnfold}$  is a retraction, identifying only Scott's two encodings of
the product bottom ( $\text{pairL Delta}$  and  $\text{pairR Delta}$ ). After the matching
 $\text{ent\_bot}$  clauses, closed elements are saturated for that kernel, so
image and preimage of  $\text{treeUnfold}$  are inverse order isomorphisms of
domains.
-/

namespace ScottModels

open Scott1982
open Scott1982.Constructive
open Scott1982.InfoSys

variable {a : Type*} [DecidableEq a] (A : InfoSys a)

/-- Official product of the tree system with itself. -/
abbrev TreeProd : InfoSys (ProdToken (treeSystem A) (treeSystem A)) :=
  productSystem (treeSystem A) (treeSystem A)

/-- Canonical section of  $\text{treeUnfold}$ . Product bottom folds to  $\text{pairL bot}$ . -/
def treeFold :
  SumToken a (ProdToken (treeSystem A) (treeSystem A)) -> TreeToken a
| .bot => .bot
| .left x => .atom x
| .right p =>
  if p.val.2 = treeBot then .pairL p.val.1 else .pairR p.val.2

theorem treeUnfold_treeFold (s : SumToken a (ProdToken (treeSystem A) (treeSystem A))) :
  treeUnfold A (treeFold A s) = s := by
  cases s with
  | bot => rfl
  | left x => rfl
  | right p =>
    dsimp [treeFold, treeUnfold]
    split_ifs with hb
    * apply congrArg SumToken.right
    * apply Subtype.ext
    * apply Prod.ext
    * rfl
    * exact hb.symm
    * have ha : p.val.1 = treeBot := by
      rcases p.property with h1 | h2
      * exact h1
      * exact False.elim (hb h2)
    * apply congrArg SumToken.right
    * apply Subtype.ext
    * apply Prod.ext
    * exact ha.symm
    * rfl

theorem treeFold_eq_or_pairBots (t : TreeToken a) :
  treeFold A (treeUnfold A t) = t \ /
  t = .pairR .bot /\ treeFold A (treeUnfold A t) = .pairL .bot := by
  cases t with
  | bot => exact Or.inl rfl
  | atom x => exact Or.inl rfl

```

```

| pairL t =>
  refine Or.inl ?_
  simp [treeFold, treeUnfold, treeBot]
| pairR t =>
  simp only [treeFold, treeUnfold, treeBot]
  split_ifs with hb
  * cases t with
    | bot => exact Or.inr <rfl, rfl>
    | atom _ | pairL _ | pairR _ => exact False.elim (nomatch hb)
  * exact Or.inl (if_neg hb)

private def unfoldInsert (t : TreeToken a) :
  Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A))) ->
  Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A))) :=
  insert (treeUnfold A t)

private instance : LeftCommutative (unfoldInsert (A := A)) :=
  <fun p q s => insert_comm' (treeUnfold A p) (treeUnfold A q) s>

/-- Choice-free image of a tree finset under `treeUnfold`. -/
def unfoldFinset (u : Finset (TreeToken a)) :
  Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A))) :=
  Multiset.foldr (unfoldInsert A) {} u.1

private theorem mem_foldr_unfold (m : Multiset (TreeToken a))
  (s : SumToken a (ProdToken (treeSystem A) (treeSystem A))) :
  s in Multiset.foldr (unfoldInsert A) {} m <->
  exists t in m, treeUnfold A t = s := by
  refine Multiset.induction_on m ?_ ?_
  * constructor
  * intro hs; exact False.elim (Finset.notMem_empty s hs)
  * rintro <_, hp, _>; exact False.elim (by cases hp)
  * intro t rest ih
  simp only [Multiset.foldr_cons, unfoldInsert, Finset.mem_insert, ih, Multiset.mem_cons]
  constructor
  * rintro (h | <q, hq, hq'>)
    * exact <t, Or.inl rfl, h.symm>
    * exact <q, Or.inr hq, hq'>
  * rintro <q, hq, hq'>
    rcases hq with rfl | hq
    * exact Or.inl hq'.symm
    * exact Or.inr <q, hq, hq'>

theorem mem_unfoldFinset {u : Finset (TreeToken a)}
  {s : SumToken a (ProdToken (treeSystem A) (treeSystem A))} :
  s in unfoldFinset A u <-> exists t in u, treeUnfold A t = s := by
  unfold unfoldFinset
  rw [mem_foldr_unfold]
  simp only [Finset.mem_def]

private def foldInsert (s : SumToken a (ProdToken (treeSystem A) (treeSystem A))) :
  Finset (TreeToken a) -> Finset (TreeToken a) :=
  insert (treeFold A s)

private instance : LeftCommutative (foldInsert (A := A)) :=
  <fun p q t => insert_comm' (treeFold A p) (treeFold A q) t>

/-- Choice-free image of an RHS finset under `treeFold`. -/
def foldFinset (Y : Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A)))) :
  Finset (TreeToken a) :=
  Multiset.foldr (foldInsert A) {} Y.1

private theorem mem_foldr_fold

```

```

(m : Multiset (SumToken a (ProdToken (treeSystem A) (treeSystem A))))
(t : TreeToken a) :
  t in Multiset.foldr (foldInsert A) {} m <->
    exists s in m, treeFold A s = t := by
refine Multiset.induction_on m ?_ ?_
* constructor
  * intro ht; exact False.elim (Finset.notMem_empty t ht)
  * rintro <_, hp, _>; exact False.elim (by cases hp)
* intro s rest ih
  simp only [Multiset.foldr_cons, foldInsert, Finset.mem_insert, ih, Multiset.mem_cons]
  constructor
  * rintro (h | <q, hq, hq'>)
    * exact <s, Or.inl rfl, h.symm>
    * exact <q, Or.inr hq, hq'>
  * rintro <q, hq, hq'>
    rcases hq with rfl | hq
    * exact Or.inl hq'.symm
    * exact Or.inr <q, hq, hq'>

theorem mem_foldFinset
  {Y : Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A)))}
  {t : TreeToken a} :
    t in foldFinset A Y <-> exists s in Y, treeFold A s = t := by
  unfold foldFinset
  rw [mem_foldr_fold]
  simp only [Finset.mem_def]

theorem unfoldFinset_foldFinset
  (Y : Finset (SumToken a (ProdToken (treeSystem A) (treeSystem A)))) :
    unfoldFinset A (foldFinset A Y) = Y := by
  ext s
  constructor
  * intro hs
    rcases (mem_unfoldFinset A).1 hs with <t, ht, hU>
    rcases (mem_foldFinset A).1 ht with <s', hs', rfl>
    rw [treeUnfold_treeFold] at hU
    rwa [ <- hU]
  * intro hs
    exact (mem_unfoldFinset A).2 <treeFold A s,
      (mem_foldFinset A).2 <s, hs, rfl>, treeUnfold_treeFold A s>

theorem lftFinset_image_unfold (u : Finset (TreeToken a)) :
  lftFinset (unfoldFinset A u) = atomFinset u := by
  ext x
  constructor
  * intro hx
    have : SumToken.left x in unfoldFinset A u := (mem_lftFinset).1 hx
    rcases (mem_unfoldFinset A).1 this with <t, ht, hU>
    cases t with
    | atom y =>
      have : y = x := SumToken.left.inj hU
      subst this
      exact mem_atomFinset.2 ht
    | bot | pairL _ | pairR _ => exact nomatch hU
  * intro hx
    exact (mem_lftFinset).2
      ((mem_unfoldFinset A).2 <.atom x, mem_atomFinset.1 hx, rfl>)

theorem rhtFinset_image_unfold_mem (u : Finset (TreeToken a))
  (y : ProdToken (treeSystem A) (treeSystem A)) :
  y in rhtFinset (unfoldFinset A u) <->
    (exists t, .pairL t in u /\ y = <(t, treeBot), Or.inr rfl>) /\
      (exists t, .pairR t in u /\ y = <(treeBot, t), Or.inl rfl>) := by

```

```

constructor
* intro hy
  have : SumToken.right y in unfoldFinset A u := (mem_rhtFinset).1 hy
  rcases (mem_unfoldFinset A).1 this with <t, ht, hU>
  cases t with
  | pairL s =>
    refine Or.inl <s, ht, ?_>
    exact SumToken.right.inj hU.symm
  | pairR s =>
    refine Or.inr <s, ht, ?_>
    exact SumToken.right.inj hU.symm
  | bot | atom _ => exact nomatch hU
* rintro (<t, ht, rfl> | <t, ht, rfl>)
* exact (mem_rhtFinset).2 ((mem_unfoldFinset A).2 <.pairL t, ht, rfl>)
* exact (mem_rhtFinset).2 ((mem_unfoldFinset A).2 <.pairR t, ht, rfl>)

theorem TreeCon_image_sumCon {u : Finset (TreeToken a)} (hu : TreeCon A u) :
  SumCon A (TreeProd A) (unfoldFinset A u) := by
  cases hu with
  | left _ hA hL hR =>
    refine Or.inl <?_, ?_>
    * rw [lftFinset_image_unfold]; exact hA
    * ext y
      constructor
      * intro hy
        rcases (rhtFinset_image_unfold_mem A u y).1 hy with <t, ht, _> | <t, ht, _>
        * exact False.elim (Finset.notMem_empty t (hL |> mem_pairLFinset.2 ht))
        * exact False.elim (Finset.notMem_empty t (hR |> mem_pairRFinset.2 ht))
      * intro hy
        exact False.elim (Finset.notMem_empty y hy)
  | right _ hA hL hR =>
    refine Or.inr <?_, ?_>
    * rw [lftFinset_image_unfold]; exact hA
    * refine <?_, ?_>
      * -- fst of the product tokens is a TreeCon set (pairL, plus bot if needed)
        have hsub :
          fstFinset (treeSystem A) (treeSystem A)
            (rhtFinset (unfoldFinset A u)) subseq
            insert .bot (pairLFinset u) := by
            intro t ht
            rcases (mem_fstFinset (treeSystem A) (treeSystem A)).1 ht with <p, hp, hb, rfl>
            rcases (rhtFinset_image_unfold_mem A u p).1 hp with <s, hs, hp'> | <s, hs, hp'>
            * subst hp'
              exact Finset.mem_insert_of_mem (mem_pairLFinset.2 hs)
            * subst hp'
              exact Finset.mem_insert_self _ _
          exact (treeSystem A).con_subset (TreeCon_insert_bot A hL) hsub
        * have hsub :
          sndFinset (treeSystem A) (treeSystem A)
            (rhtFinset (unfoldFinset A u)) subseq
            insert .bot (pairRFinset u) := by
            intro t ht
            rcases (mem_sndFinset (treeSystem A) (treeSystem A)).1 ht with <p, hp, ha, rfl>
            rcases (rhtFinset_image_unfold_mem A u p).1 hp with <s, hs, hp'> | <s, hs, hp'>
            * subst hp'
              exact Finset.mem_insert_self _ _
            * subst hp'
              exact Finset.mem_insert_of_mem (mem_pairRFinset.2 hs)
          exact (treeSystem A).con_subset (TreeCon_insert_bot A hR) hsub

theorem SumCon_image_treeCon {u : Finset (TreeToken a)}
  (hu : SumCon A (TreeProd A) (unfoldFinset A u)) : TreeCon A u := by
  rcases hu with <hL, hr> | <hl, hR>

```

```

* refine TreeCon.left _ (by rwa [lftFinset_image_unfold] at hL) ?_ ?_
* ext t
  constructor
* intro ht
  have : (<(t, treeBot), Or.inr rfl> : ProdToken (treeSystem A) (treeSystem A)) in
    rhtFinset (unfoldFinset A u) :=
      (rhtFinset_image_unfold_mem A u _).2 (Or.inl <t, mem_pairLFinset.1 ht, rfl>)
  rw [hr] at this
  exact False.elim (Finset.notMem_empty _ this)
* intro ht
  exact False.elim (Finset.notMem_empty t ht)
* ext t
  constructor
* intro ht
  have : (<(treeBot, t), Or.inl rfl> : ProdToken (treeSystem A) (treeSystem A)) in
    rhtFinset (unfoldFinset A u) :=
      (rhtFinset_image_unfold_mem A u _).2 (Or.inr <t, mem_pairRFinset.1 ht, rfl>)
  rw [hr] at this
  exact False.elim (Finset.notMem_empty _ this)
* intro ht
  exact False.elim (Finset.notMem_empty t ht)
* refine TreeCon.right _ (by rwa [lftFinset_image_unfold] at hL) ?_ ?_
* have hsub : pairLFinset u subseteq
  fstFinset (treeSystem A) (treeSystem A)
  (rhtFinset (unfoldFinset A u)) := by
  intro t ht
  exact (mem_fstFinset (treeSystem A) (treeSystem A)).2
    <<(t, treeBot), Or.inr rfl>,
    (rhtFinset_image_unfold_mem A u _).2 (Or.inl <t, mem_pairLFinset.1 ht, rfl>),
    rfl, rfl>
  exact (treeSystem A).con_subset hR.1 hsub
* have hsub : pairRFinset u subseteq
  sndFinset (treeSystem A) (treeSystem A)
  (rhtFinset (unfoldFinset A u)) := by
  intro t ht
  exact (mem_sndFinset (treeSystem A) (treeSystem A)).2
    <<(treeBot, t), Or.inl rfl>,
    (rhtFinset_image_unfold_mem A u _).2 (Or.inr <t, mem_pairRFinset.1 ht, rfl>),
    rfl, rfl>
  exact (treeSystem A).con_subset hR.2 hsub

theorem TreeEnt_sumEnt {u : Finset (TreeToken a)} {p : TreeToken a}
  (h : TreeEnt A u p) :
  SumEnt A (TreeProd A) (unfoldFinset A u) (treeUnfold A p) := by
  have hCon : SumCon A (TreeProd A) (unfoldFinset A u) :=
    TreeCon_image_sumCon A h.1
  refine <hCon, ?_>
  cases p with
  | bot => exact trivial
  | atom x =>
    refine <?_, ?_>
    * rw [lftFinset_image_unfold]; exact h.2.1
    * rw [lftFinset_image_unfold]; exact h.2.2
  | pairL t =>
    have hr : rhtFinset (unfoldFinset A u) != {} := by
      rcases h.2.2.2 with <hne, _> | <_, hR>
    * obtain <s, hs> := Finset.nonempty_of_ne_empty hne
      exact Finset.ne_empty_of_mem
        ((rhtFinset_image_unfold_mem A u _).2
          (Or.inl <s, mem_pairLFinset.1 hs, rfl>))
    * obtain <s, hs> := Finset.nonempty_of_ne_empty hR
      exact Finset.ne_empty_of_mem
        ((rhtFinset_image_unfold_mem A u _).2

```

```

      (Or.inr <s, mem_pairRFinset.1 hs, rfl>))
refine <hr, ?_>
refine <hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2), ?_, ?_>
* intro
  -- T.Ent fst t
  have hsub : pairLFinset u subseteq
    fstFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) := by
    intro s hs
    exact (mem_fstFinset (treeSystem A) (treeSystem A)).2
      <<(s, treeBot), Or.inr rfl>,
      (rhtFinset_image_unfold_mem A u _).2 (Or.inl <s, mem_pairLFinset.1 hs, rfl>),
      rfl, rfl>
  have hFstCon : fstFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) in (treeSystem A).Con :=
    hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2.1)
  rcases h.2.2.2.2 with <hne, hPay> | <rfl, _>
  * exact (treeSystem A).ent_trans hFstCon h.2.2.1
    (fun y hy => (treeSystem A).ent_refl hFstCon (hsub hy)) <h.2.2.1, hPay>
  * exact (treeSystem A).ent_bot hFstCon
* intro ht
  have hSndCon : sndFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) in (treeSystem A).Con :=
    hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2.2)
  exact (treeSystem A).ent_bot hSndCon

| pairR t =>
  have hr : rhtFinset (unfoldFinset A u) != {} := by
    rcases h.2.2.2.2 with <hne, _> | <_, hL>
  * obtain <s, hs> := Finset.nonempty_of_ne_empty hne
    exact Finset.ne_empty_of_mem
      ((rhtFinset_image_unfold_mem A u _).2
        (Or.inr <s, mem_pairRFinset.1 hs, rfl>))
  * obtain <s, hs> := Finset.nonempty_of_ne_empty hL
    exact Finset.ne_empty_of_mem
      ((rhtFinset_image_unfold_mem A u _).2
        (Or.inl <s, mem_pairLFinset.1 hs, rfl>))
refine <hr, ?_>
refine <hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2), ?_, ?_>
* intro
  have hFstCon : fstFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) in (treeSystem A).Con :=
    hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2.1)
  exact (treeSystem A).ent_bot hFstCon
* intro
  have hsub : pairRFinset u subseteq
    sndFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) := by
    intro s hs
    exact (mem_sndFinset (treeSystem A) (treeSystem A)).2
      <<(treeBot, s), Or.inl rfl>,
      (rhtFinset_image_unfold_mem A u _).2 (Or.inr <s, mem_pairRFinset.1 hs, rfl>),
      rfl, rfl>
  have hSndCon : sndFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) in (treeSystem A).Con :=
    hCon.elim (fun hL => False.elim (hr hL.2)) (fun hR => hR.2.2)
  rcases h.2.2.2.2 with <hne, hPay> | <rfl, _>
  * exact (treeSystem A).ent_trans hSndCon h.2.2.1
    (fun y hy => (treeSystem A).ent_refl hSndCon (hsub hy)) <h.2.2.1, hPay>
  * exact (treeSystem A).ent_bot hSndCon

theorem SumEnt_treeFold {u : Finset (TreeToken a)}
  {s : SumToken a (ProdToken (treeSystem A) (treeSystem A))}

```

```

(h : SumEnt A (TreeProd A) (unfoldFinset A u) s) :
TreeEnt A u (treeFold A s) := by
have hu : TreeCon A u := SumCon_image_treeCon A h.1
refine <hu, ?_>
cases s with
| bot => exact trivial
| left x =>
  dsimp [treeFold]
  refine <?_, ?_>
  * rw [← lftFinset_image_unfold A u]; exact h.2.1
  * rw [← lftFinset_image_unfold A u]; exact h.2.2
| right p =>
  dsimp [treeFold]
  have hAt : atomFinset u = {} := by

```

Lean 4 source (lines 401–644)

```

have h1 := (lftFinset_image_unfold A u)
have : lftFinset (unfoldFinset A u) = {} :=
  SumCon_lft_empty_of_rht_nonempty A (TreeProd A) h.1 h.2.1
  rwa [h1] at this
split_ifs with hb
* refine <hAt, TreeCon_pairL_of_right A hu hAt, TreeCon_pairR_of_right A hu hAt, ?_>
  have hEnt := h.2.2
  -- ProdEnt: Ent fst p.1 (since p.2 = bot)
  have hFst : (treeSystem A).Ent
    (fstFinset (treeSystem A) (treeSystem A)
     (rhtFinset (unfoldFinset A u))) p.val.1 :=
    hEnt.2.1 hb
  have hsub :
    fstFinset (treeSystem A) (treeSystem A)
      (rhtFinset (unfoldFinset A u)) subseteq
      insert .bot (pairLFinset u) := by
    intro t ht
    rcases (mem_fstFinset (treeSystem A) (treeSystem A)).1 ht with <q, hq, _, rfl>
    rcases (rhtFinset_image_unfold_mem A u q).1 hq with <w, hw, hq'> | <w, hw, hq'>
    * subst hq'; exact Finset.mem_insert_of_mem (mem_pairLFinset.2 hw)
    * subst hq'; exact Finset.mem_insert_self _ _
  -- Transfer Ent along the subset into pairL U {bot}
  have hIns : TreeCon A (insert .bot (pairLFinset u)) :=
    TreeCon_insert_bot A (TreeCon_pairL_of_right A hu hAt)
  have hPay : TreeEntPayload A (insert .bot (pairLFinset u)) p.val.1 :=
    ((treeSystem A).ent_trans hIns
     (SumCon_rht_con_of_rht_nonempty A (TreeProd A) h.1 h.2.1).1
     (fun y hy => (treeSystem A).ent_refl hIns (hsub hy)) hFst).2
  obtain <y, hy> := Finset.nonempty_of_ne_empty h.2.1
  rcases (rhtFinset_image_unfold_mem A u y).1 hy with <w, hwL, _> | <w, hwR, _>
  * refine Or.inl <Finset.ne_empty_of_mem (mem_pairLFinset.2 hwL), ?_>
    exact ((treeSystem A).ent_trans (TreeCon_pairL_of_right A hu hAt) hIns
      (fun z hz => by
        rw [Finset.mem_insert] at hz
        rcases hz with rfl | hz
        * exact (treeSystem A).ent_bot (TreeCon_pairL_of_right A hu hAt)
        * exact (treeSystem A).ent_refl (TreeCon_pairL_of_right A hu hAt) hz)
      <hIns, hPay>).2
  * if ht : p.val.1 = .bot then
    exact Or.inr <ht, Finset.ne_empty_of_mem (mem_pairRFinset.2 hwR)>
  else
    refine Or.inl <?_, ?_>
    * intro hempty
    have hEq : insert (.bot : TreeToken a) (pairLFinset u) = {.bot} := by
      rw [hempty]; rfl
    exact ht (TreeEntPayload_of_no_atoms_no_pairs A
      atomFinset_singleton_bot pairLFinset_singleton_bot

```

```

    pairRFinset_singleton_bot (hEq |> hPay))
* exact ((treeSystem A).ent_trans (TreeCon_pairL_of_right A hu hAt) hIns
  (fun z hz => by
    rw [Finset.mem_insert] at hz
    rcases hz with rfl | hz
    * exact (treeSystem A).ent_bot (TreeCon_pairL_of_right A hu hAt)
    * exact (treeSystem A).ent_refl (TreeCon_pairL_of_right A hu hAt) hz)
  <hIns, hPay>).2
* -- pairR p.2
refine <hAt, TreeCon_pairR_of_right A hu hAt, TreeCon_pairL_of_right A hu hAt, ?_>
have ha : p.val.1 = treeBot := by
  rcases p.property with h1 | h2
  * exact h1
  * exact False.elim (hb h2)
have hEnt := h.2.2
have hSnd : (treeSystem A).Ent
  (sndFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u))) p.val.2 :=
  hEnt.2.2 ha
have hIns : TreeCon A (insert .bot (pairRFinset u)) :=
  TreeCon_insert_bot A (TreeCon_pairR_of_right A hu hAt)
have hsub :
  sndFinset (treeSystem A) (treeSystem A)
    (rhtFinset (unfoldFinset A u)) subseteq
  insert .bot (pairRFinset u) := by
  intro t ht
  rcases (mem_sndFinset (treeSystem A) (treeSystem A)).1 ht with <q, hq, _, rfl>
  rcases (rhtFinset_image_unfold_mem A u q).1 hq with <w, hw, hq'> | <w, hw, hq'>
  * subst hq'; exact Finset.mem_insert_self _ _
  * subst hq'; exact Finset.mem_insert_of_mem (mem_pairRFinset.2 hw)
have hPay : TreeEntPayload A (insert .bot (pairRFinset u)) p.val.2 :=
  ((treeSystem A).ent_trans hIns
    (SumCon_rht_con_of_rht_nonempty A (TreeProd A) h.1 h.2.1).2
    (fun y hy => (treeSystem A).ent_refl hIns (hsub hy)) hSnd).2
refine Or.inl <?_, ?_>
* intro hempty
have hEq : insert (.bot : TreeToken a) (pairRFinset u) = {.bot} := by
  rw [hempty]; rfl
have ht : p.val.2 = .bot :=
  TreeEntPayload_of_no_atoms_no_pairs A
    atomFinset_singleton_bot pairLFinset_singleton_bot
    pairRFinset_singleton_bot (hEq |> hPay)
exact hb (ht |> rfl)
* exact ((treeSystem A).ent_trans (TreeCon_pairR_of_right A hu hAt) hIns
  (fun z hz => by
    rw [Finset.mem_insert] at hz
    rcases hz with rfl | hz
    * exact (treeSystem A).ent_bot (TreeCon_pairR_of_right A hu hAt)
    * exact (treeSystem A).ent_refl (TreeCon_pairR_of_right A hu hAt) hz)
  <hIns, hPay>).2

theorem TreeEnt_pairL_bot_of_pairs {u : Finset (TreeToken a)}
  (hu : TreeCon A u) (h : pairLFinset u != {} \ / pairRFinset u != {}):
  TreeEnt A u (.pairL .bot) :=
  <hu, TreeCon_atoms_empty_of_right A hu h,
    TreeCon_pairL_of_right A hu (TreeCon_atoms_empty_of_right A hu h),
    TreeCon_pairR_of_right A hu (TreeCon_atoms_empty_of_right A hu h),
    h.elim (fun hL => Or.inl <hL, trivial>) (fun hR => Or.inr <rfl, hR>)>

theorem TreeEnt_pairR_bot_of_pairs {u : Finset (TreeToken a)}
  (hu : TreeCon A u) (h : pairLFinset u != {} \ / pairRFinset u != {}):
  TreeEnt A u (.pairR .bot) :=
  <hu, TreeCon_atoms_empty_of_right A hu h,

```

```

TreeCon_pairR_of_right A hu (TreeCon_atoms_empty_of_right A hu h),
TreeCon_pairL_of_right A hu (TreeCon_atoms_empty_of_right A hu h),
h.elim (fun hL => Or.inr <rfl, hL>) (fun hR => Or.inl <hR, trivial>))

theorem mem_closed_pair_bots (x : (treeSystem A).Element) :
  .pairL .bot in x.carrier <-> .pairR .bot in x.carrier := by
  constructor
  * intro h
    have hsub : (({TreeToken.pairL treeBot} : Finset (TreeToken a)) : Set _) subseq
      x.carrier := by
      intro a ha
      have : a = .pairL treeBot := Finset.mem_singleton.mp (Finset.mem_coe.1 ha)
      subst this; exact h
    exact x.closed {TreeToken.pairL treeBot} (.pairR treeBot) hsub
      (TreeEnt_pairR_bot_of_pairs A (x.consistent _ hsub)
        (Or.inl (Finset.ne_empty_of_mem (by
          rw [pairLFinset_singleton_pairL]; exact Finset.mem_singleton_self _))))))
  * intro h
    have hsub : (({TreeToken.pairR treeBot} : Finset (TreeToken a)) : Set _) subseq
      x.carrier := by
      intro a ha
      have : a = .pairR treeBot := Finset.mem_singleton.mp (Finset.mem_coe.1 ha)
      subst this; exact h
    exact x.closed {TreeToken.pairR treeBot} (.pairL treeBot) hsub
      (TreeEnt_pairL_bot_of_pairs A (x.consistent _ hsub)
        (Or.inr (Finset.ne_empty_of_mem (by
          rw [pairRFinset_singleton_pairR]; exact Finset.mem_singleton_self _))))))

theorem saturated_of_closed (x : (treeSystem A).Element) {t : TreeToken a}
  (ht : t in x.carrier) : treeFold A (treeUnfold A t) in x.carrier := by
  rcases treeFold_eq_or_pairBots A t with h | <rfl, hf>
  * rwa [h]
  * rw [hf]
  exact (mem_closed_pair_bots A x).2 ht

/-- Image of a tree element under `treeUnfold` is an official RHS element. -/
def unfoldElement (x : (treeSystem A).Element) : (treeRhs A).Element where
  carrier := treeUnfold A '' x.carrier
  consistent := by
    intro Y hY
    have hFold : (foldFinset A Y) subseq x.carrier := by
      intro t ht
      rcases (mem_foldFinset A).1 (Finset.mem_coe.1 ht) with <s, hs, rfl>
      have hs' : s in treeUnfold A '' x.carrier := hY (Finset.mem_coe.2 hs)
      rcases hs' with <t', ht', hU>
      have : treeFold A (treeUnfold A t') in x.carrier := saturated_of_closed A x ht'
      rwa [hU] at this
    have hCon : TreeCon A (foldFinset A Y) := x.consistent _ hFold
    have : SumCon A (TreeProd A) (unfoldFinset A (foldFinset A Y)) :=
      TreeCon_image_sumCon A hCon
    rwa [unfoldFinset_foldFinset] at this
  closed := by
    intro Y s hY hEnt
    have hFold : (foldFinset A Y) subseq x.carrier := by
      intro t ht
      rcases (mem_foldFinset A).1 (Finset.mem_coe.1 ht) with <s', hs', rfl>
      have hs'' : s' in treeUnfold A '' x.carrier := hY (Finset.mem_coe.2 hs')
      rcases hs'' with <t', ht', hU>
      have : treeFold A (treeUnfold A t') in x.carrier := saturated_of_closed A x ht'
      rwa [hU] at this
    have hEnt' : SumEnt A (TreeProd A)
      (unfoldFinset A (foldFinset A Y)) s := by
      rwa [unfoldFinset_foldFinset]

```

```

have hT : TreeEnt A (foldFinset A Y) (treeFold A s) :=
  SumEnt_treeFold A hEnt'
have ht : treeFold A s in x.carrier :=
  x.closed _ _ hFold hT
refine <treeFold A s, ht, treeUnfold_treeFold A s>

/-- Preimage of an official RHS element under `treeUnfold` is a tree element. -/
def foldElement (y : (treeRhs A).Element) : (treeSystem A).Element where
  carrier := treeUnfold A ^-1' y.carrier
  consistent := by
    intro Y hY
    have him : (unfoldFinset A Y) subseq y.carrier := by
      intro s hs
      rcases (mem_unfoldFinset A).1 (Finset.mem_coe.1 hs) with <t, ht, rfl>
      exact hY (Finset.mem_coe.2 ht)
    have : SumCon A (TreeProd A) (unfoldFinset A Y) := y.consistent _ him
    exact SumCon_image_treeCon A this
  closed := by
    intro Y p hY hEnt
    have him : (unfoldFinset A Y) subseq y.carrier := by
      intro s hs
      rcases (mem_unfoldFinset A).1 (Finset.mem_coe.1 hs) with <t, ht, rfl>
      exact hY (Finset.mem_coe.2 ht)
    have hS : SumEnt A (TreeProd A) (unfoldFinset A Y) (treeUnfold A p) :=
      TreeEnt_sumEnt A hEnt
    exact y.closed _ _ him hS

theorem unfoldElement_foldElement (y : (treeRhs A).Element) :
  unfoldElement A (foldElement A y) = y := by
  refine le_antisymm ?_ ?_
  * intro s hs
    rcases hs with <t, ht, rfl>
    exact ht
  * intro s hs
    refine <treeFold A s, ?_, treeUnfold_treeFold A s>
    change treeUnfold A (treeFold A s) in y.carrier
    rwa [treeUnfold_treeFold]

theorem foldElement_unfoldElement (x : (treeSystem A).Element) :
  foldElement A (unfoldElement A x) = x := by
  refine le_antisymm ?_ ?_
  * intro t ht
    rcases ht with <t', ht', hU>
    have : treeFold A (treeUnfold A t') in x.carrier := saturated_of_closed A x ht'
    have hft : treeFold A (treeUnfold A t) in x.carrier := by
      rwa [hU] at this
    rcases treeFold_eq_or_pairBots A t with h | <rfl, hf>
    * rwa [h] at hft
    * exact (mem_closed_pair_bots A x).1 (by rwa [hf] at hft)
  * intro t ht
    exact <t, ht, rfl>

/-- **Factoid 8.1 at domain level.** `|T| ?o |A + (T x T)|` via `treeUnfold`. -/
def treeDomainIso : (treeSystem A).Element ?o (treeRhs A).Element where
  toFun := unfoldElement A
  invFun := foldElement A
  left_inv := foldElement_unfoldElement A
  right_inv := unfoldElement_foldElement A
  map_rel_iff' := by
    intro x_1 x_2
    constructor
    * intro h t ht
      have : treeUnfold A t in (unfoldElement A x_1).carrier := <t, ht, rfl>

```

```

    have : treeUnfold A t in (unfoldElement A x_2).carrier := h this
    have : t in (foldElement A (unfoldElement A x_2)).carrier := this
    rwa [foldElement_unfoldElement] at this
  * intro h s hs
    rcases hs with <t, ht, rfl>
    exact <t, h ht, rfl>

end ScottModels

```

A.11 ScottModels/WorkedExampleSEExpr.lean

173 lines.

Lean 4 source

```

/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import Mathlib.Order.Hom.Basic
import Mathlib.Order.Hom.WithTopBot
import Mathlib.Data.Sum.Order
import Mathlib.Order.Ideal
import Scott1982.Factoid24
import Scott1982.Factoid81
import Scott1982.Proposition54
import ScottModels.InfoSysToNeighborhood
import ScottModels.InfoSysToIdealCompletion
import ScottModels.PresentationDomains
import ScottModels.InfoSysConstructions
import ScottModels.ScottMapBridge
import ScottModels.SexDomainEquation

/-!
# Worked example -- S-expression / tree domain `T ~ A + (T x T)`

Scott 1982 Factoid 8.1 (`treeSystem`) over the Nat lower-bound atom system
(Factoid 2.4), walked through this package's bridges: information system ->
neighbourhood filters -> ideal completion, plus identity approximable maps as
Scott-continuous maps (Factoid 4.6).
-/

namespace ScottModels

open Scott1982
open Scott1982.Constructive
open Scott1982.InfoSys
open Scott1982.InfoSys.ApproximableMap
open Order

/-! ## Atom system (Factoid 2.4, named) -/

/-- Scott's Nat lower-bound information system (Factoid 2.4), as a named definition. -/
def lowerBoundSystem : InfoSys Nat where
  bot := 0
  Con := Set.univ
  Ent := Factoid24.lowerBoundEnt
  con_subset := by
    intro u v _ _
    exact Set.mem_univ v

```

```

con_sing := by
  intro _
  exact Set.mem_univ _
ent_con := by
  intro u a _
  exact Set.mem_univ _
ent_bot := by
  intro u _
  exact Or.inl rfl
ent_refl := by
  intro u a _ ha
  exact Or.inr <a, ha, le_refl>
ent_trans := by
  intro u v c _ _ hvEnt huEnt
  rcases huEnt with rfl | <n, hn, hcn>
  * exact Or.inl rfl
  * rcases hvEnt n hn with rfl | <k, hk, hnk>
    * exact Or.inl (Nat.le_zero.mp hcn)
    * exact Or.inr <k, hk, le_trans hcn hnk>

/#! ## S-expression system -/

/-- Tree / S-expression information system over lower-bound atoms. -/
abbrev SexSys : InfoSys (TreeToken Nat) :=
  treeSystem lowerBoundSystem

/-- Official right-hand side `A + (T x T)`. -/
abbrev SexRhs : InfoSys (SumToken Nat (ProdToken SexSys SexSys)) :=
  treeRhs lowerBoundSystem

theorem sexRhs_eq_sum_product :
  SexRhs = sumSystem lowerBoundSystem (productSystem SexSys SexSys) :=
  treeRhs_eq_sum_product lowerBoundSystem

/-- Unfolding tokens into the sum-of-product carrier (Factoid 8.1). -/
theorem sexUnfold_atom (n : Nat) :
  treeUnfold lowerBoundSystem (.atom n) = SumToken.left n :=
  treeUnfold_atom lowerBoundSystem n

/#! ## Concrete finite elements -/

/-- Closure of a singleton atom token `{atom n}`. -/
noncomputable def sexAtom (n : Nat) : SexSys.Element :=
  SexSys.closure {TreeToken.atom n} (SexSys.con_sing _)

theorem sexAtom_mem_self (n : Nat) : TreeToken.atom n in (sexAtom n).carrier :=
  SexSys.subset_closure (SexSys.con_sing _) (Finset.mem_singleton_self _)

/#! ## 1982 -> 1980: basic-open neighbourhood filters -/

/-- `|T| ?o` filters of basic opens `[u]`. -/
noncomputable abbrev sexNeighborhoodIso :
  SexSys.Element ?o (InfoSysToNeighborhood.toNeighborhoodSystem SexSys).Element :=
  InfoSysToNeighborhood.domainOrderIso SexSys

/#! ## 1982 -> ideal completion -/

/-- `|T| ?o Ideal (FiniteElement T)`. -/
noncomputable abbrev sexIdealIso :
  SexSys.Element ?o Ideal (InfoSysToIdealCompletion.FiniteElement SexSys) :=
  InfoSysToIdealCompletion.domainOrderIso SexSys

/-- Neighbourhood filters ? ideals of finite elements (constructive triangle). -/

```

```

noncomputable abbrev sexNeighborhoodIdealIso :=
  neighborhood_ideal_iso SexSys

/ -! ## Domain equation at the level of domains (1982 constructions) -/

/--
Semantic factor:  $|A + (T \times T)| \circ \text{WithBot} (|A| \circ (|T| \times |T|))$  via the generic
product and separated-sum isos. Not the fixed-point equation itself.
-/
noncomputable def sexRhsSemanticIso :
  WithBot (lowerBoundSystem.Element  $\circ$  (SexSys.Element  $\times$  SexSys.Element))  $\circ$ 
  SexRhs.Element :=
  let ?Prod := InfoSysConstructions.productDomainIso SexSys SexSys
  let ?Sum := InfoSysConstructions.sumDomainIso lowerBoundSystem (productSystem SexSys SexSys)
  let mid :=
    (OrderIso.refl lowerBoundSystem.Element).sumCongr ?Prod |>.withBotCongr
  mid.trans ?Sum

/--
**S-expression fixed-point equation.  $|T| \circ |A + (T \times T)|$ , the order
isomorphism of  $\text{SexSys.Element}$  with  $\text{SexRhs.Element}$  induced by
 $\text{treeUnfold} / \text{treeFold} (\text{treeDomainIso})$ .
-/
noncomputable def sexDomainEquationIso :
  SexSys.Element  $\circ$  SexRhs.Element :=
  treeDomainIso lowerBoundSystem

theorem sexDomainEquationIso_unfold (x : SexSys.Element) :
  (sexDomainEquationIso x).carrier =
    treeUnfold lowerBoundSystem 'x'.carrier :=
  rfl

theorem sexDomainEquationIso_fold (y : SexRhs.Element) :
  (sexDomainEquationIso.symm y).carrier =
    treeUnfold lowerBoundSystem  $\sim 1$ ' y.carrier :=
  rfl

theorem exists_sexNeighborhoodIso :
  Nonempty (SexSys.Element  $\circ$ 
    (InfoSysToNeighborhood.toNeighborhoodSystem SexSys).Element) :=
  <sexNeighborhoodIso>

theorem exists_sexIdealIso :
  Nonempty (SexSys.Element  $\circ$ 
    Ideal (InfoSysToIdealCompletion.FiniteElement SexSys)) :=
  <sexIdealIso>

theorem exists_sexDomainEquationIso :
  Nonempty (SexSys.Element  $\circ$  SexRhs.Element) :=
  <sexDomainEquationIso>

/ -! ## Morphisms: identity is Scott-continuous (Factoid 4.6) -/

/-- Identity approximable map on  $T$ , as a Scott-continuous endomap. -/
noncomputable abbrev sexIdScottContinuous : ScottContinuous SexSys SexSys :=
  (approximableMap_scottContinuous_equiv SexSys SexSys) (idMap SexSys)

theorem sexId_toElement (x : SexSys.Element) :
  sexIdScottContinuous.toFun x = x :=
  idMap_toElement SexSys x

end ScottModels

```

A.12 ScottModels/Equivalence.lean

41 lines.

Lean 4 source

```
/-
Copyright (c) 2026 Lars Warren Ericson. All rights reserved.
Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/scott\_models
-/

import ScottModels.NeighborhoodToInfoSys
import ScottModels.InfoSysToNeighborhood
import ScottModels.ContinuousLatticeToNeighborhood
import ScottModels.InfoSysToIdealCompletion
import ScottModels.IdealCompletionToContinuousLattice
import ScottModels.PresentationDomains
import ScottModels.InfoSysConstructions
import ScottModels.ScottMapBridge

/-!
# Equivalence theorems

Bridges between Scott's 1972 continuous-lattice presentation, 1980 neighbourhood systems
(PRG-19), and 1982 information systems. Sibling packages are finished; this module holds
the cross-presentation maps. See `arxiv.md` / `HANDOFF.md`.

## Status

* `neighborhoodSystem_to_infoSys` -- **done** (`NeighborhoodToInfoSys.lean`)
* `infoSys_to_neighborhoodSystem` -- **done** (`InfoSysToNeighborhood.lean`)
* `continuousLattice_to_neighborhoodSystem` -- **done** (`ContinuousLatticeToNeighborhood.lean`);
  `D ?o RoundFilter`, not raw `|?|`
* `infoSys_to_idealCompletion` -- **done** (`InfoSysToIdealCompletion.lean`;
  `|A| ?o Ideal (FiniteElement A)`)
* `idealCompletion_to_continuousLattice` -- **done** (`IdealCompletionToContinuousLattice.lean`);
  algebraic complete lattice => `IsContinuousLattice`; classical `<<`
* `presentation_domains_equiv` -- **done** (`PresentationDomains.lean`;
  `D ?o RoundFilter ?o RoundInfoSysElement` via `wayBelowNbhdBasis`)
* `infoSys_constructions_equiv` -- **done** (`InfoSysConstructions.lean` +
  `ScottMapBridge.lean`; 1982 product/sum/function-space domain isos;
  `ApproximableMap ?o ScottContinuous`; `ScottMap` conjugates along round presentation)
* worked example -- **done** (`WorkedExampleSEExpr.lean`; Factoid 8.1 trees through
  neighbourhood / ideal bridges + domain-equation iso)
-/
```